

ملخص لاب كنترول

إعداد: حلمي نزال

الطبعة الثالثة

كيفية دراسة اللاب

1. الدراسة على المانيوال.

2. حل تكليفات المانيوال مع التطبيق على الماتلاب.

3. تسجيل ملاحظاتك على التجربة.

4. الدراسة على الدوسية.

5. حل الأمثلة والواجبات.

6. حل الامتحانات الموجودة في الدوسية.

*ملاحظة: التجربة (7) صفحات (50 – 62) بتندرس على ال PDF

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Part (A): First Order System

Part (B): Second Order System

Experiment (8)

Experiment (9)

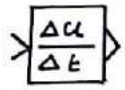
Final Exam in Experiments (9, 8, 7, 6)

Experiment(1)

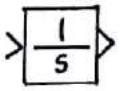
Control tutorials for Simulink

* البلوكات المستخدمة في تجارب المختبر موجودة في Library ← Simulink

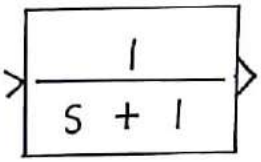
a. continuous



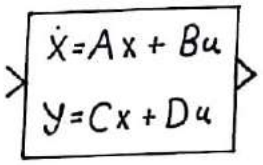
derivative



integrator

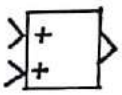


transfer fcn

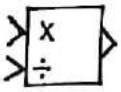


state-space

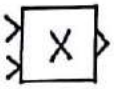
b. math operations



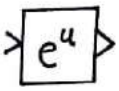
add



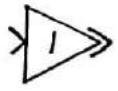
divide



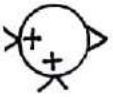
Product



Math Function



gain



sum

c. signal routing


mux

d. sinks

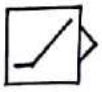

scope

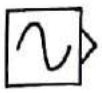
e. sources



constant


step


ramp


sine wave

* كيفية تعديل بلوك Transfer Fcn ؟

Ex1: Modify the tranfer function block to give the following function, $G(s) = \frac{13s + 8}{2s^3 + 16s^2 + 17}$

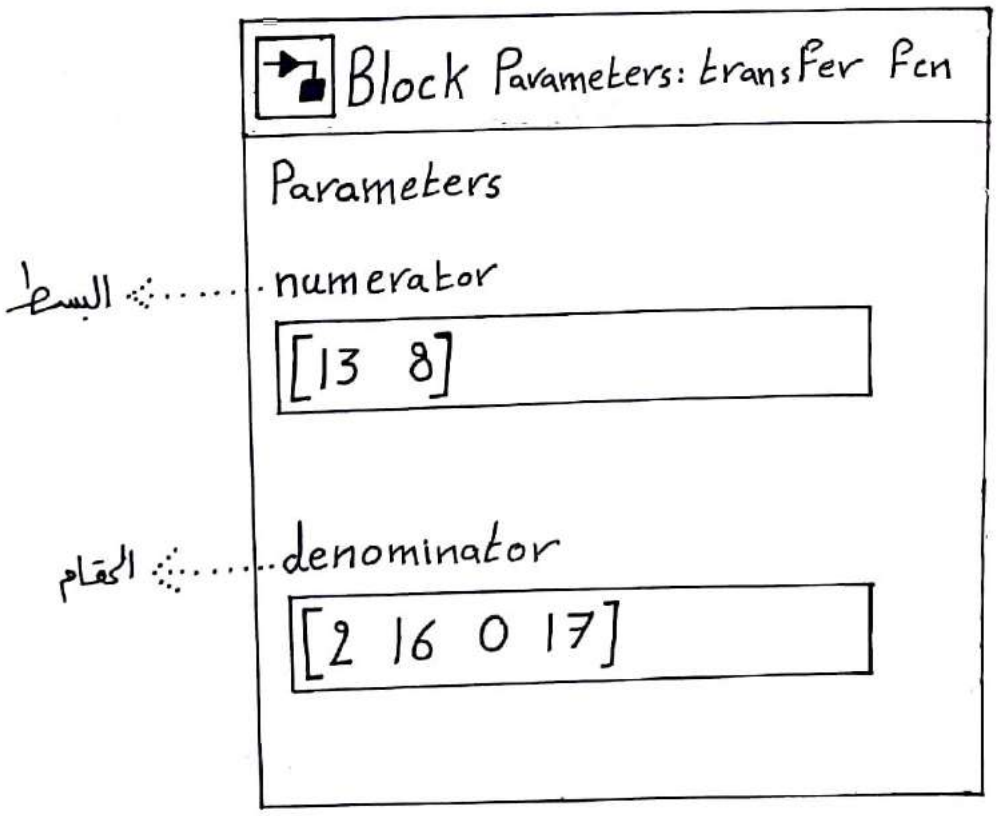
1. تحديد معاملات كل من البسط والمقام.

البسط ← $13s^1 + 8s^0$ ← المعاملات 13 8

المقام ← $2s^3 + 16s^2 + 0s^1 + 17s^0$ ← المعاملات 2 16 0 17

2. ندخل معاملات البسط في مصفوفة البسط ومعاملات المقام في مصفوفة المقام مع مراعاة وجود مسافة (space) بين كل معامل والآخر، حيث تتم كتابة معاملات كل من البسط والمقام داخل [ ]

• solution



* كيفية رسم simulation diagram لاقتران معين:

$$y(t) = a_0 + b.t + c.\sin\omega t + d\int y(t).dt + e.\frac{dy(t)}{dt}$$

a ⟶ constant [a]

b, c, d, e ⟶ gain

sinωt ⟶ sine wave

t ⟶ ramp

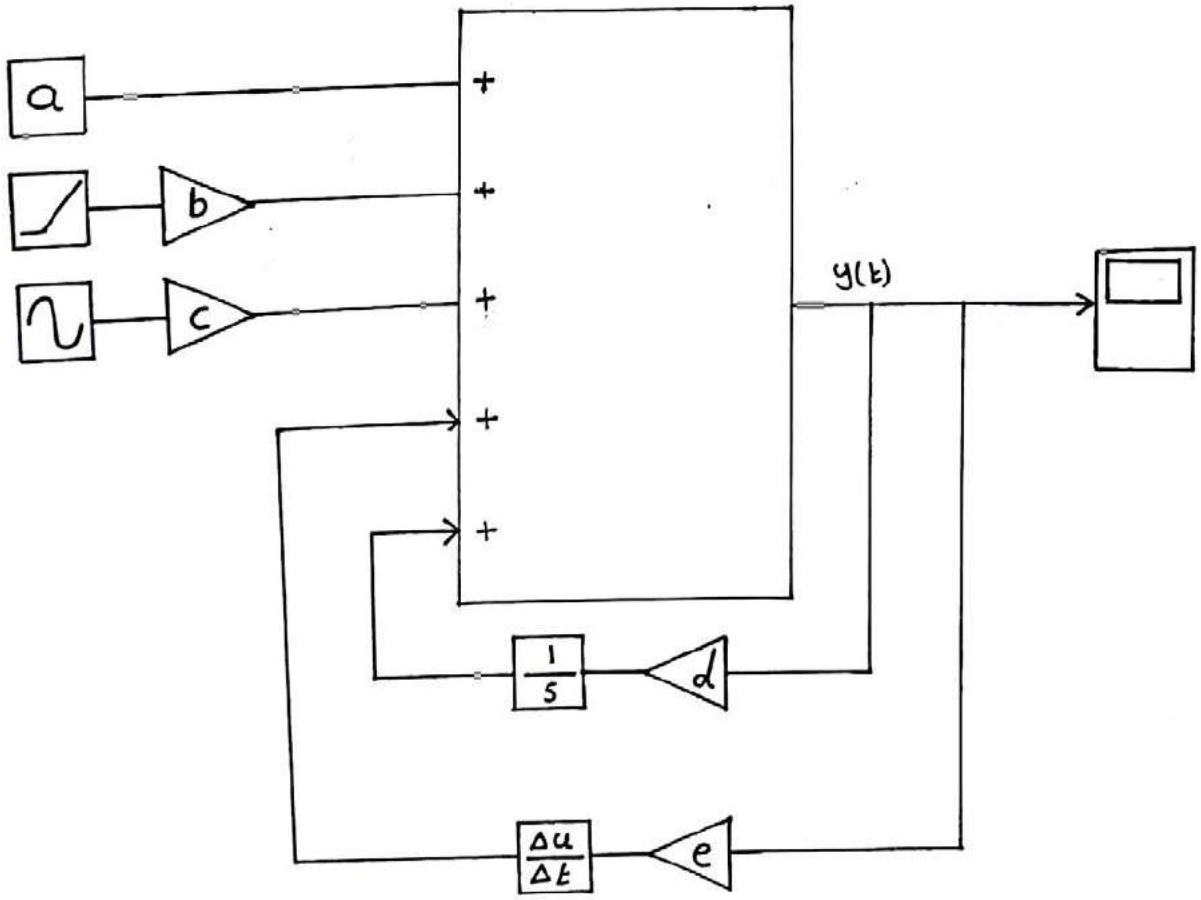
$\int$ ⟶ integrator [$\frac{1}{s}$]

$\frac{d}{dt}$ ⟶ derivative [$\frac{\Delta u}{\Delta t}$]

+ ⟶ add

y(t) ⟶ Output

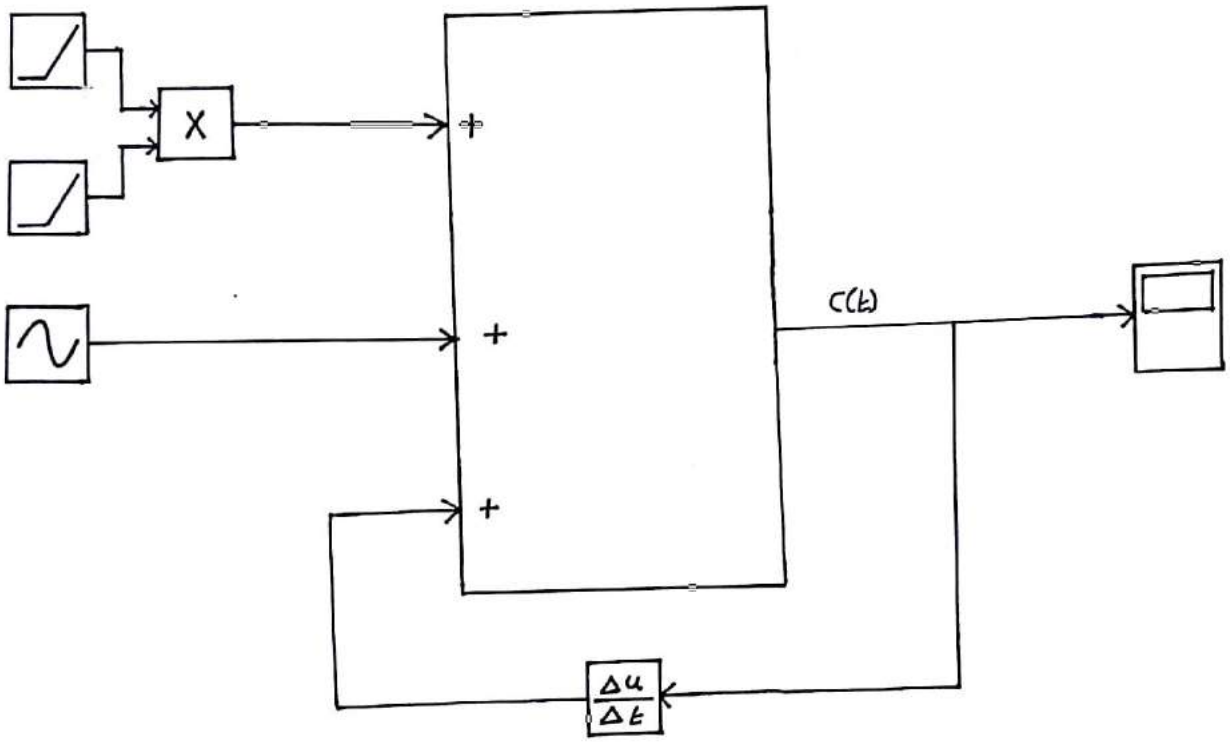
* يتم رسم simulation diagram كالتالي:



Ex_2 : Draw the simulation diagram for the following function:

$$c(t) = t^2 + \sin 30t + \frac{dc(t)}{dt}$$

- solution



H.w$_1$: write the numerator and denominator parameters as in Simulink for the following transfer function:

$$G(s) = \frac{8s + 12}{s^3 + 9s^2 + 9s + 3}$$

numerator [ ] , denominator [ ]

H.w$_2$: Draw the simulation diagram for the following function:

$$y(t) = 3 + 0.2t - 5\int y(t).dt$$

Laplace transform

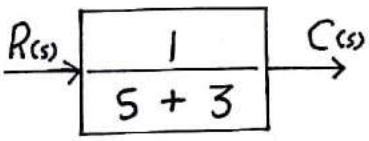
* $\mathcal{L}[f(t)] = F(s)$

time domain → frequency domain

$$f(t) \underset{\mathcal{L}^{-1}}{\overset{\mathcal{L}}{\rightleftarrows}} F(s)$$

step	a	$\frac{a}{s}$
ramp	at	$\frac{a}{s^2}$
exponential	e^{-at}	$\frac{1}{s+a}$
derivation	$\frac{d^n(f(t))}{dt^n}$	$s^n.F(s)$
integration	$\int f(t).dt$	$\frac{1}{s}.F(s)$

Ex: Consider the system below, find $c(t)$ when $r(t) = 2$



• Solution:

$r(t) = 2 \longrightarrow R(s) = \frac{2}{s}$

كسور جزئية

$$\frac{C(s)}{R(s)} = \frac{1}{s+3} \longrightarrow C(s) = \frac{1}{s+3} \cdot R(s) = \frac{1}{s+3} \cdot \frac{2}{s} = \frac{2}{s(s+3)} = \frac{A}{s} + \frac{B}{s+3}$$

$$2 = A(s+3) + Bs$$

$s = 0 \rightarrow 2 = 3A \rightarrow A = \frac{2}{3}$

$s = -3 \rightarrow 2 = -3B \rightarrow B = \frac{-2}{3}$

$$C(s) = \frac{2/3}{s} + \frac{-2/3}{s+3}$$

$c(t) = \mathcal{L}^{-1}[C(s)]$

$$c(t) = \mathcal{L}\left[\frac{2/3}{s} + \frac{-2/3}{s+3}\right]$$

$$= \frac{2}{3} - \frac{2}{3} \cdot e^{-3t}$$

$$c(t) = \frac{2}{3}(1 - e^{-3t})$$

Experiment (2)
Solving differential equations

a. Solving linear differential equation by integration

* الخطوات:

1. نجعل أعلى مشتقة موضع القانون.
2. نقسم المعادلة على معامل أعلى مشتقة (قيمة معامل أعلى مشتقة يجب أن تساوي 1).
3. نرسم simulation diagram.

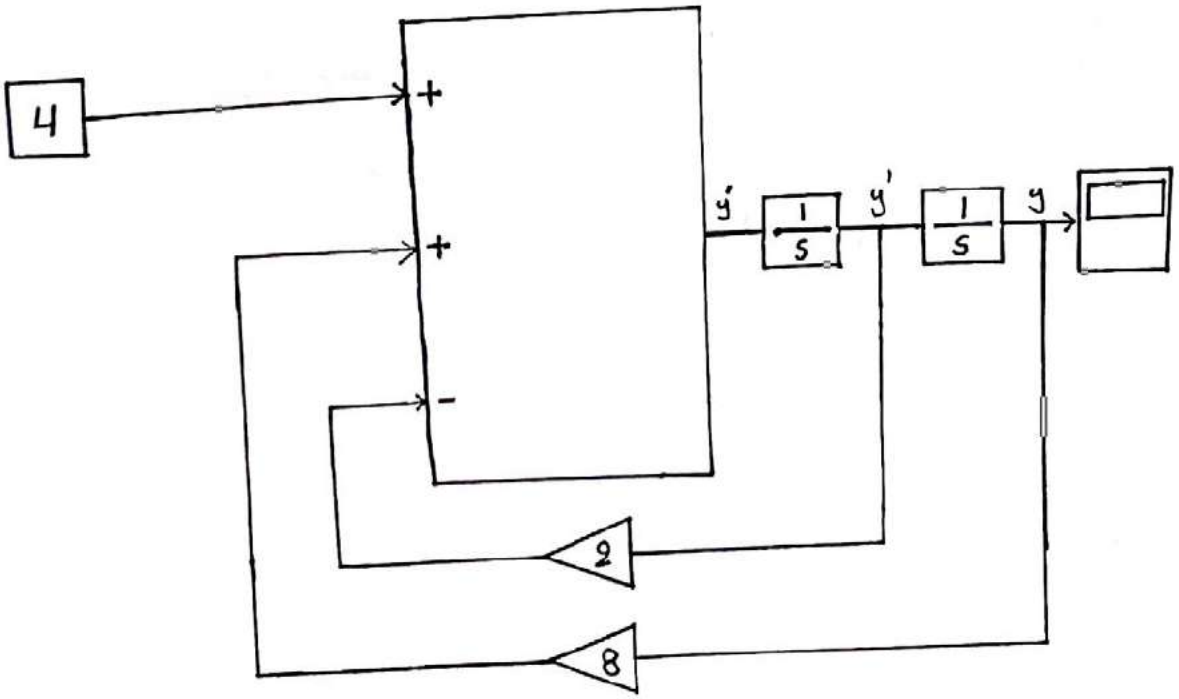
Ex1: Solve the following differential equation by integration and draw the simulation diagram:

$$2y'' + 4y' - 16y = 8$$

- Solution:

$$2y'' + 4y' - 16y = 8 \xrightarrow[\text{موضع القانون}]{y'' \text{ نجعل}} 2y'' = 8 - 4y' + 16y \xrightarrow[y'' \text{ معامل}]{\text{القسمة على}} y'' = 4 - 2y' + 8y$$

- Simulation diagram:



b. solving set of differential equation by integration

* الخطوات :

- نفس الخطوات السابقة ولكن بأخذ كل معادلة على حدة.

Ex$_2$: Solve the following differential equations by integration and draw the simulation diagram:

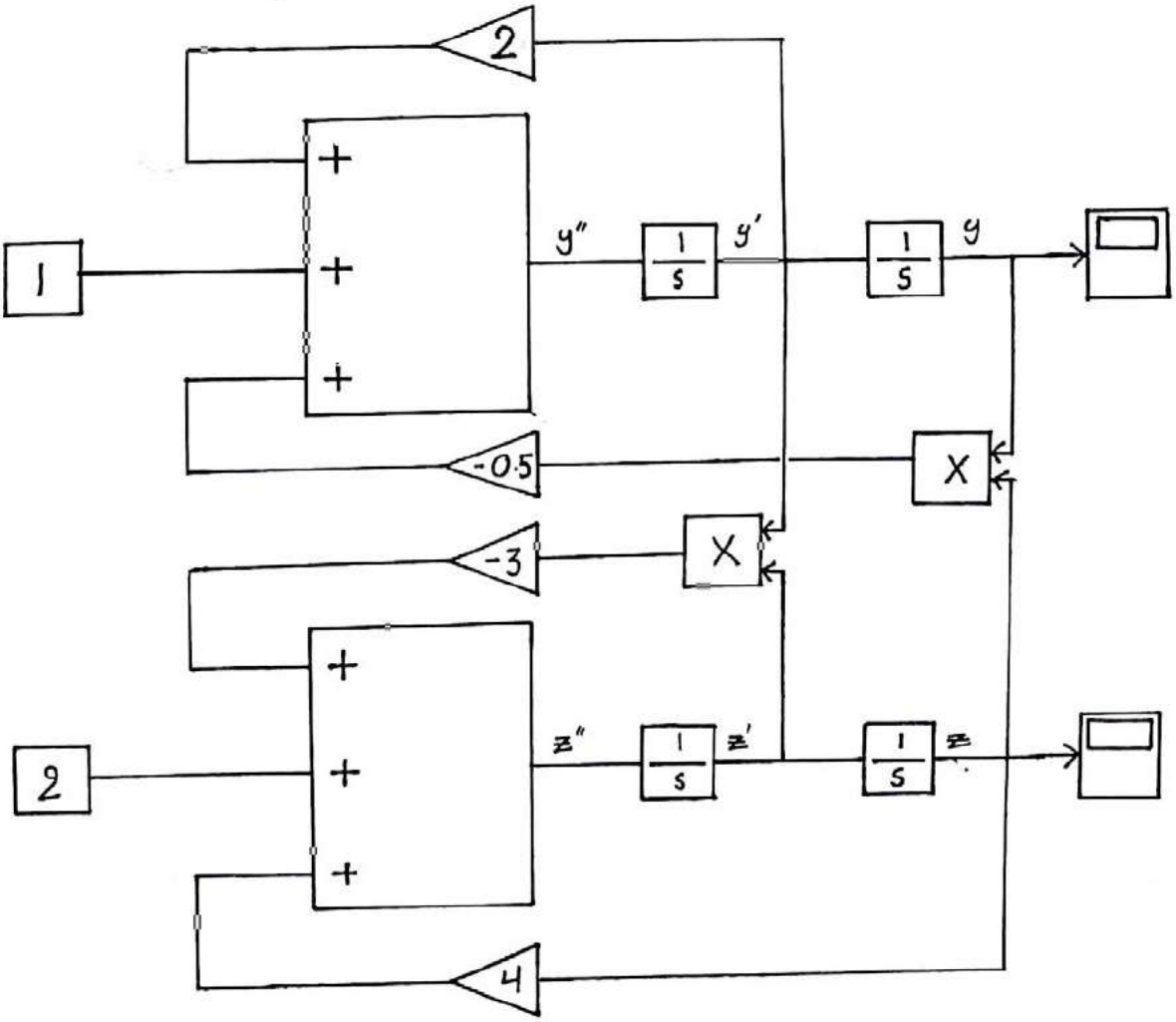
$$-2y'' + 4y' - zy = -2$$
$$3z'' + 9y'z' - 12z = 6$$

- Solution:

$-2y'' + 4y' - zy = -2$ $\xrightarrow[\text{موضع القانون}]{\text{نجعل } y''}$ $-2y'' = -2 - 4y' + zy$ $\xrightarrow[\text{معامل } y'']{\text{نقسم على}}$ $y'' = 1 + 2y' - 0.5zy$

$3z'' + 9y'z' - 12z = 6$ $\xrightarrow[\text{موضع القانون}]{\text{نجعل } z''}$ $3z'' = 6 - 9y'z' + 12z$ $\xrightarrow[\text{معامل } z'']{\text{نقسم على}}$ $z'' = 2 - 3y'z' + 4z$

- simulation diagram:



c. transfer function approach

* الخطوات:

1. نطبق لابلاس على المعادلة بحيث أن $y^{(n)}(t) \xrightarrow{\mathcal{L}} s^n.Y(s)$ ، $x(t) \xrightarrow{\mathcal{L}} X(s)$.
2. نأخذ $Y(s)$ عامل مشترك.
3. نقسم المعادلة على $X(s)$.
4. نجعل المقدار $\frac{Y(s)}{X(s)}$ موضع القانون.
5. نرسم simulation diagram.

Ex$_3$: Find the transfer function for the following equation and draw the simulation diagram, assume $r(t)=2$

$$y''' + 3y'' - 6y' + 17y = r(t)$$

• solution:

$$y''' + 3y'' - 6y' + 17y = r(t)$$

1. تطبيق $\mathcal{L}$ على المعادلة ←

$$s^3Y(s) + 3s^2Y(s) - 6sY(s) + 17Y(s) = R(s)$$

2. أخذ $Y(s)$ عامل مشترك ←

$$Y(s).(s^3+3s^2-6s+17) = R(s)$$

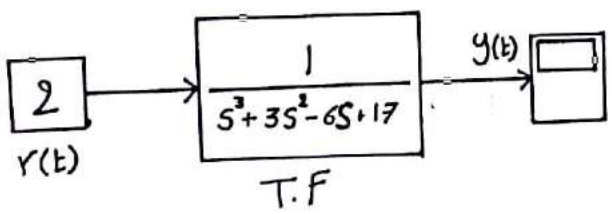
3. قسمة المعادلة على $X(s)$ ←

$$\frac{Y(s)}{R(s)}(s^3+3s^2-6s+17) = 1$$

4. جعل $\frac{Y(s)}{X(s)}$ موضع القانون ←

$$\frac{Y(s)}{R(s)} = \frac{1}{s^3+3s^2-6s+17}$$

• simulation diagram: ← 5



*لاحظ: استخدمنا القيمة $(r(t)=2)$ في الرسم وليس في إيجاد $\frac{Y(s)}{R(s)}$ (transfer function).

d. State space approach

$x'(t) = Ax(t) + Bu(t)$ ⟶ input equation

$y(t) = Cx(t) + Du(t)$ ⟶ output equation

* الخطوات:

1. نجعل أعلى مشتقة موضع القانون.
2. نقسم المعادلة على معامل أعلى مشتقة (قيمة معامل أعلى مشتقة يجب أن تساوي 1).
3. نفرض أن كل مشتقة في المعادلة (ما عدا أعلى مشتقة) تساوي State Variable.
4. نشتق الفرض السابق لينتج لدينا State equations.
5. نعوض قيم State Variables في المعادلات الناتجة (State equations).
6. نحول المعادلات إلى صيغة مصفوفات.

* ملاحظات:

- y^n, y و n: المشتقة: $y^{(n)}$, $y^n \neq y^{(n)}$
- y: y مرفوع للقوة n.
- y ⟵ المشتقة الصفرية.

* حجم المصفوفات يعتمد على:

1. n: عدد State variables.
2. r: عدد المدخلات (inputs).
3. m: عدد المخرجات (outputs).

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}_{n \times n}$$

$$B = \begin{bmatrix} b_{11} & \cdots & b_{1r} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{nr} \end{bmatrix}_{n \times r}$$

$$C = \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & & \vdots \\ c_{m1} & \cdots & c_{mn} \end{bmatrix}_{m \times n}$$

$$D = \begin{bmatrix} d_{11} & \cdots & d_{1r} \\ \vdots & & \vdots \\ d_{m1} & \cdots & d_{mr} \end{bmatrix}_{m \times r}$$

m ← عدد الصفوف

r → عدد الأعمدة

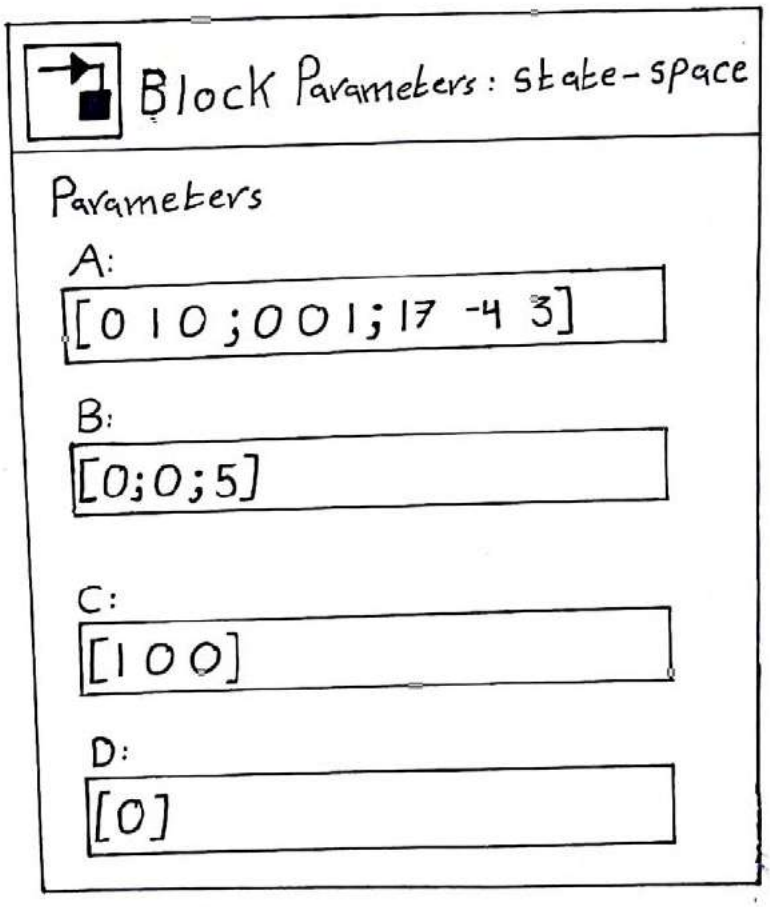
* كيفية تعبئة قيم المصفوفات (A, B, C, D) داخل بلوك state-space في Simulink :

1. كتابة القيم بين [ ] .
2. تعبئة قيم كل صف على حدة، من اليسار إلى اليمين مع مراعاة وجود مسافة (space) بين القيمة والأخرى.
3. عند الانتقال إلى تعبئة قيم الصف التالي ندخل الرمز ;

Ex4: the values of matrices (A, B, C, D) are shown below, enter their as in Simulink block modification.

$$A=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 17 & -4 & 3\end{bmatrix} \quad B=\begin{bmatrix}0\\ 0\\ 5\end{bmatrix} \quad C=\begin{bmatrix}1 & 0 & 0\end{bmatrix} \quad D=\begin{bmatrix}0\end{bmatrix}$$

- Solution:



Ex5: Find the state space representation for the following differential equation and determine the values of matrices (A, B, C, D).

$$2y'' + 6y' - 8y = 4u$$

- solution:

$$2y'' + 6y' - 8y = 4u \xrightarrow{\text{نجعل "y" موضع القانون}} 2y'' = 4u - 6y' + 8y \xrightarrow{\text{نقسم على معامل "y"}} y'' = 2u - 3y' + 4y$$

- state variables

$$X_1 = y \longrightarrow X_1' = y' = X_2$$

$$X_2 = y' \longrightarrow X_2' = y'' = 2u - 3y' + 4y$$
$$= 2u - 3X_2 + 4X_1$$

- state space representation

$$\begin{bmatrix} X_1' \\ X_2' \end{bmatrix} = \overset{X_1}{\overset{X_2}{\begin{bmatrix} 0 & 1 \\ 4 & -3 \end{bmatrix}}} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u(t)$$

$$[y] = \overset{X_1}{\overset{X_2}{\begin{bmatrix} 1 & 0 \end{bmatrix}}} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + [0]\, u(t)$$

- matrices:

$$A = \underset{2\times 2}{\begin{bmatrix} 0 & 1 \\ 4 & -3 \end{bmatrix}} \quad B = \underset{2\times 1}{\begin{bmatrix} 0 \\ 2 \end{bmatrix}} \quad C = \underset{1\times 2}{\begin{bmatrix} 1 & 0 \end{bmatrix}} \quad D = \underset{1\times 1}{[0]}$$

- عدد state variables ← $n = 2$
- عدد المدخلات (inputs) ← $r = 1$
- عدد المخرجات (outputs) ← $m = 1$

Ex6: Find the state space representation following differential equation and determine the values of matrices (A, B, C, D).

$$y'' + 9y' + 2y = u_1 + 2u_2$$

- solution:

$$y'' + 9y' + 2y = u_1 + 2u_2 \xrightarrow{\text{نجعل } y'' \text{ موضع القانون}} y'' = u_1 + 2u_2 - 9y' - 2y$$

- state variables

$$X_1 = y \longrightarrow X_1' = y' = X_2$$

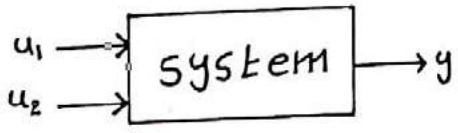
$$X_2 = y' \longrightarrow X_2' = y'' = u_1 + 2u_2 - 9y' - 2y$$
$$= u_1 + 2u_2 - 9X_2 - 2X_1$$

- state space representation

$$\begin{bmatrix} X_1' \\ X_2' \end{bmatrix} = \overset{\begin{matrix} X_1 & X_2 \end{matrix}}{\begin{bmatrix} 0 & 1 \\ -2 & -9 \end{bmatrix}} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \overset{\begin{matrix} u_1 & u_2 \end{matrix}}{\begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$[y] = \overset{\begin{matrix} X_1 & X_2 \end{matrix}}{\begin{bmatrix} 1 & 0 \end{bmatrix}} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \overset{\begin{matrix} u_1 & u_2 \end{matrix}}{\begin{bmatrix} 0 & 0 \end{bmatrix}} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$A = \underset{2\times2}{\begin{bmatrix} 0 & 0 \\ -2 & -9 \end{bmatrix}} \quad B = \underset{2\times2}{\begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}} \quad C = \underset{1\times2}{\begin{bmatrix} 1 & 0 \end{bmatrix}} \quad D = \underset{1\times2}{\begin{bmatrix} 0 & 0 \end{bmatrix}}$$



- عدد state variables ← $n = 2$
- عدد المدخلات (inputs) ← $r = 2$
- عدد المخرجات (outputs) ← $m = 1$

Ex7: Find the state space representation for the following differential equations

$$y_2' + 2y_2 + 7y_1 = 4u_1 + 3u_2$$
$$y_1'' + 9y_1' + y_1' + 2y_1 + 3y_2 = u_1 + 2u_2$$

• solution:

$$y_2' + 2y_2 + 7y_1 = 4u_1 + 3u_2 \xrightarrow{\text{نجعل } y_2' \text{ موضوع القانون}} y_2' = 4u_1 + 3u_2 - 2y_2 - 7y_1$$

$$y_1'' + 9y_1' + y_2' + 2y_1 + 3y_2 = u_1 + 2u_2 \xrightarrow[y_2']{\text{نعوض قيمة}} y_1'' + 9y_1' + (4u_1 + 3u_2 - 2y_2 - 7y_1) + 2y_1 + 3y_2 = u_1 + 2u_2$$

$$y_1'' + 9y_1' - 5y_1 + y_2 = -3u_1 - u_2 \xrightarrow{\text{نجعل } y_1'' \text{ موضوع القانون}} y_1'' = -3u_1 - u_2 - 9y_1' + 5y_1 - y_2$$

• state variables

$$X_1 = y_1 \longrightarrow X_1' = y_1' = X_2$$

$$X_2 = y_1' \longrightarrow X_2' = y_1'' = -3u_1 - u_2 - 9y_1' + 5y_1 - y_2$$
$$= -3u_1 - u_2 - 9X_2 + 5X_1 - X_3$$

$$X_3 = y_2 \longrightarrow X_3' = y_2' = 4u_1 + 3u_2 - 2y_2 - 7y_1$$
$$= 4u_1 + 3u_2 - 2X_3 - 7X_1$$

• state equations

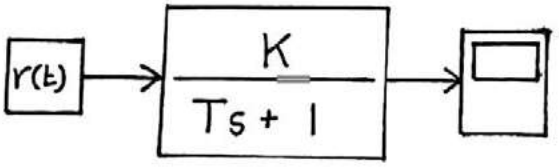
• state space representation

$$\begin{bmatrix} X_1' \\ X_2' \\ X_3' \end{bmatrix} = \overset{\begin{matrix} X_1 & X_2 & X_3 \end{matrix}}{\begin{bmatrix} 0 & 1 & 0 \\ 5 & -9 & -1 \\ -7 & 0 & -2 \end{bmatrix}} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + \overset{\begin{matrix} u_1 & u_2 \end{matrix}}{\begin{bmatrix} 0 & 0 \\ -3 & -1 \\ 4 & 3 \end{bmatrix}} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \overset{\begin{matrix} X_1 & X_2 & X_3 \end{matrix}}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + \overset{\begin{matrix} u_1 & u_2 \end{matrix}}{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Experiment (3)
Control system with feedback

* First order system



r(t): input ⟶ constant = step 

K: gain

T: time constant

c(t): output ⟶ $c(t) = O_{ss}(1 - e^{-t/T})$

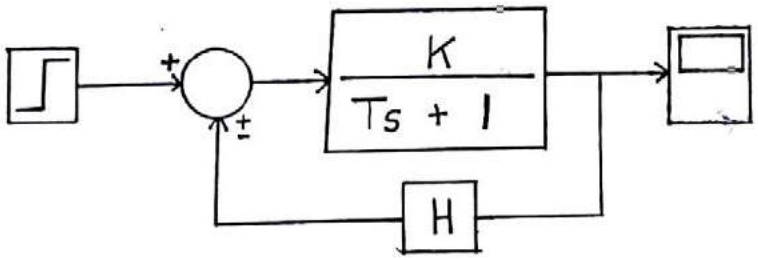
* استجابة النظام بمرور الزمن (time response):

1. في الفترة ($0 < t < t_{ss}$), تحدث الاستجابة اللحظية للنظام (transient response) تتغير فيها إشارة المخرج بشكل exponential ($-e^{-t/T}$).

2. في الفترة ($t_{ss} < t < \infty$), تحدث الإستجابة النهائية للنظام (forced response) أو ما يعرف بـ (steady state response) تثبت عندها إشارة المخرج عند القيمة (O_{ss}).

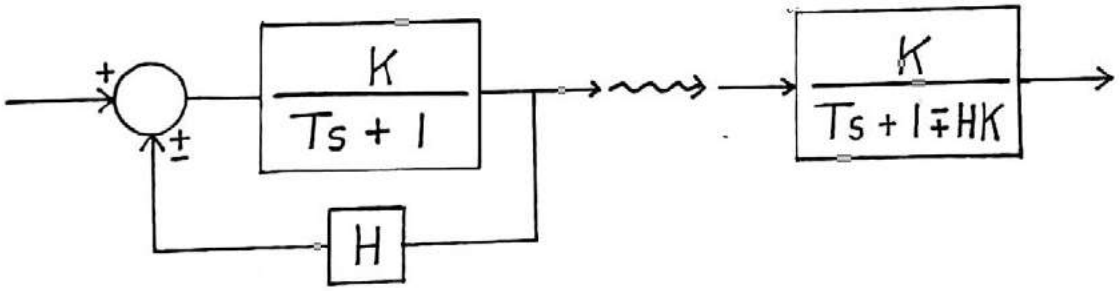
- t_{ss} : زمن الاستقرار (settling time) أو ما يعرف بـ (steady state time).
- O_{ss} : القيمة النهائية لإشارة المخرج (steady state value).
- T : ثابت الزمن (time constant).

*First order system with feedback and step input



* خطوات دراسة أثر إضافة Feedback على النظام :

1. التخلص من Feedback loop



2. نجعل القيمة $\frac{K}{Ts + \square}$ تساوي ١

إما بالقسمة (في حالة Fixed Feedback).

أو بأخذ s عامل مشترك (في حالة Flexible Feedback).

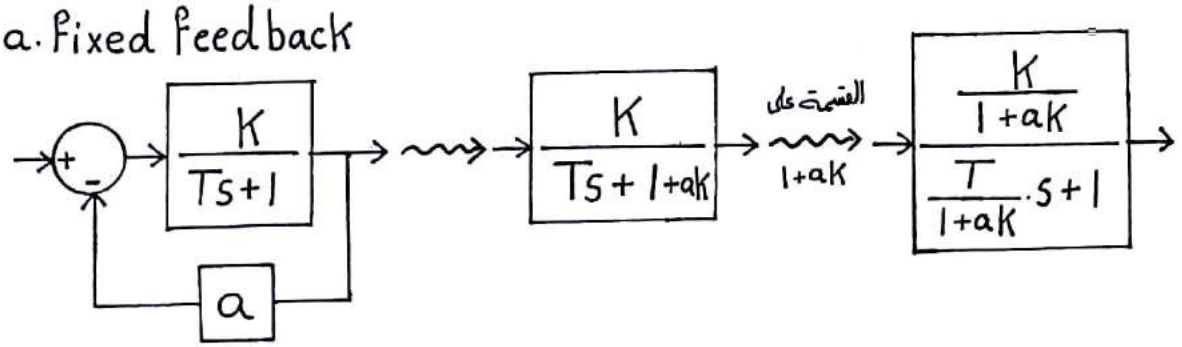
- Fixed Feedback ⟶ Feedback = gain

- Flexible Feedback ⟶ Feedback = gain & derivative

  ↳ $s = \frac{\Delta u}{\Delta t}$

1. Negative feedback

a. Fixed feedback



* أثر إضافة (Negative Fixed feedback) على كل من:

1. gain:

قبل ← gain = K

بعد ← gain = $\frac{K}{1+aK}$

* (قبل) gain > (بعد) gain

* gain ← (decreased)

2. time constant:

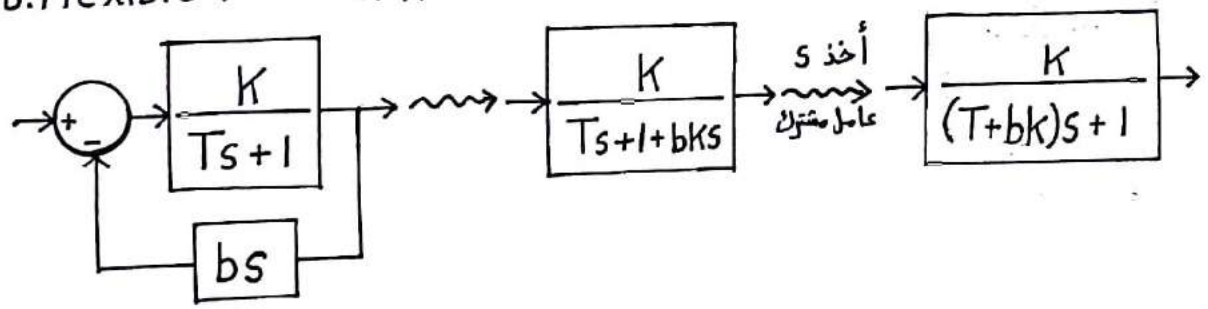
قبل ← time constant = T

بعد ← time constant = $\frac{T}{1+aK}$

* (قبل) time constant > (بعد) time constant

* time constant ← (decreased)

b. Flexible feedback



* أثر إضافة (Negative Flexible feedback) على كل من:

1. gain:

قبل ← gain = K

بعد ← gain = K

* (قبل) gain = (بعد) gain

* gain ← (no change)

2. time constant:

قبل ← time constant = T

بعد ← time constant = (T+bK)

* (قبل) time constant < (بعد) time constant

* time constant ← (increased)

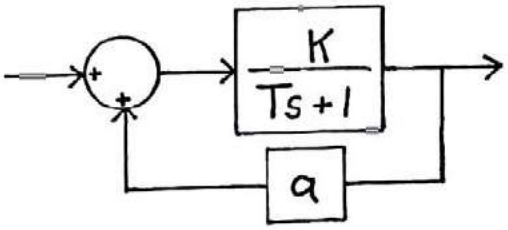
2. Positive feedback

* أثر إضافة Postive Feedback على النظام ← يجعل النظام غير مستقر (unstable).

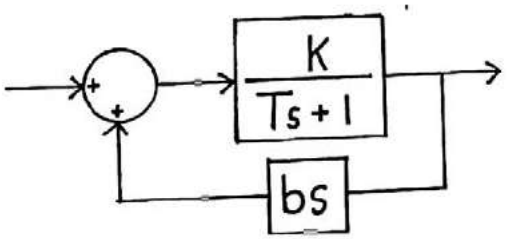
* الحالات التي يبقى فيها النظام مستقر (stable) بعد إضافة Positive feedback :

1. في حالة Fixed feedback , عندما يكون $0 < 1 - aK$ ←

2. في حالة Flexible feedback, عندما يكون $0 < T - bK$ ←

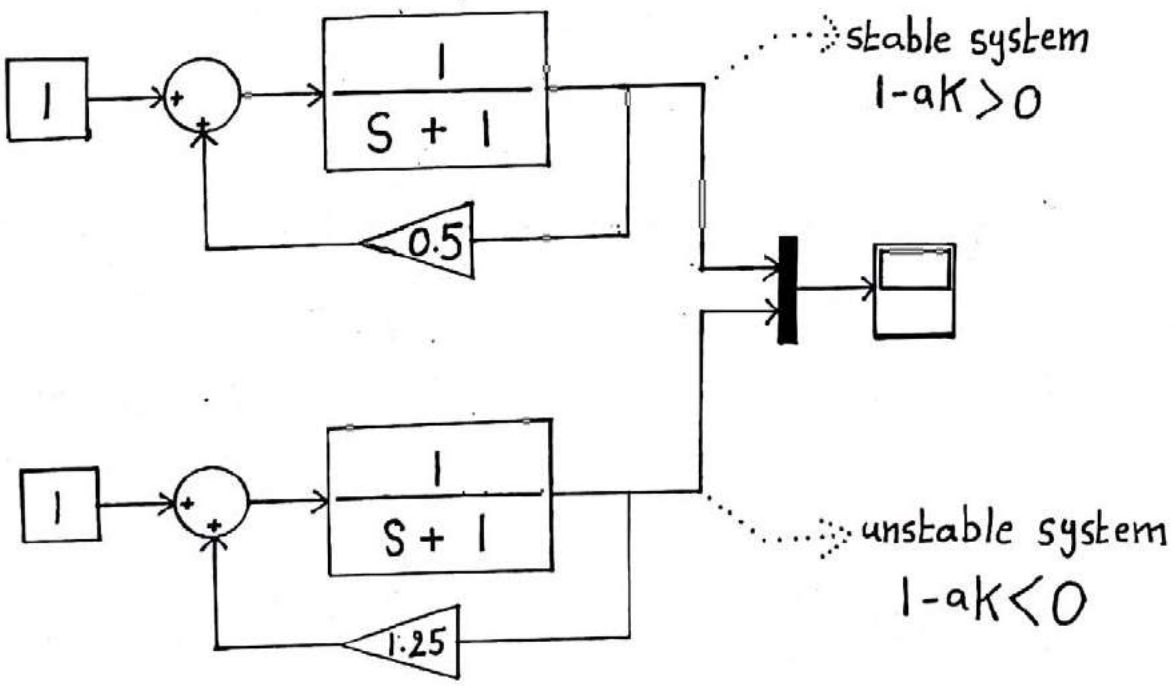


Positive Fixed feedback

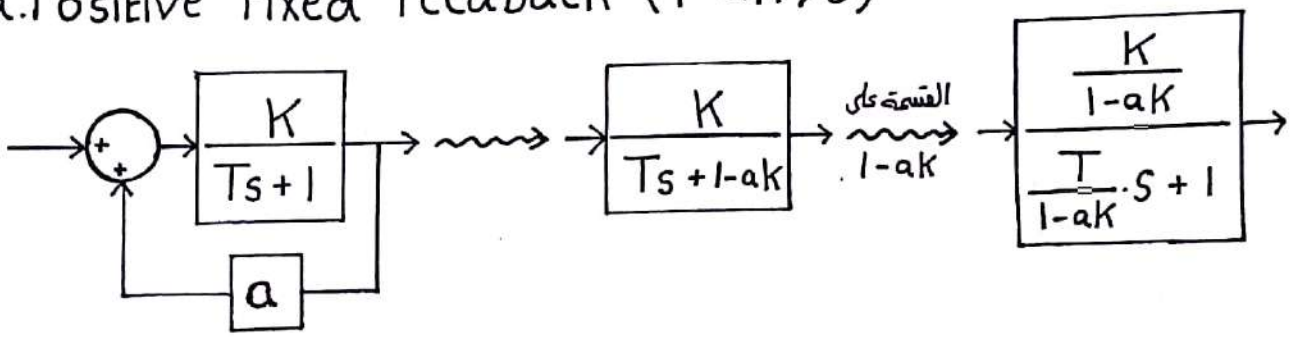


Positive Flexible feedback

* واجب : مثّل الأنظمة التالية على برنامج (matlab simulink) وقارن الإشارات الناتجة على Scope :



a. Positive fixed feedback ($1-aK>0$)



* أثر إضافة (Positive fixed feedback) على كل من:

1. gain:

قبل ← gain = K

بعد ← gain = $\frac{K}{1-aK}$

(قبل) gain < (بعد) gain *

(increased) ← gain *

2. time constant:

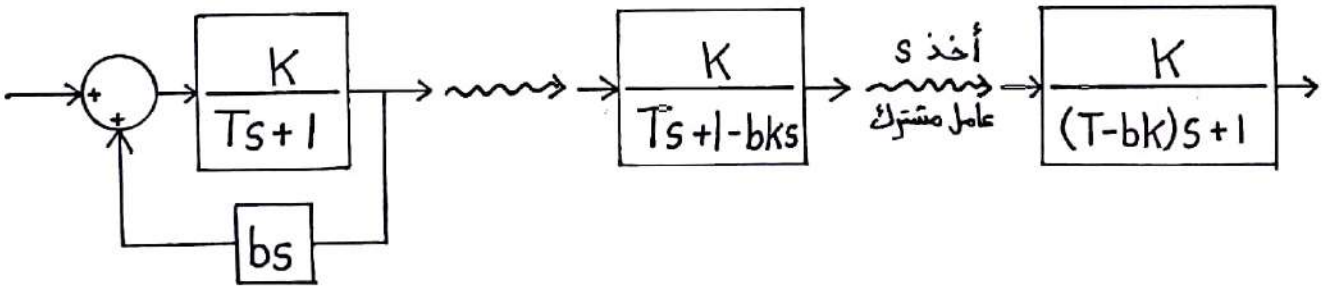
قبل ← time constant = T

بعد ← time constant = $\frac{T}{1-aK}$

(قبل) time constant < (بعد) time constant *

(increased) ← time constant *

b. Positive flexible feedback ($T-bK>0$)



* أثر إضافة (Positive flexible feedback) على كل من:

1. gain:

قبل ← gain = K

بعد ← gain = K

(قبل) gain (بعد) gain *

(no change) ← gain *

2. time constant:

قبل ← time constant = T

بعد ← time constant = (T-bK)

(قبل) time constant > (بعد) time constant *

(decreased) ← time constant *.

* خطوات رسم إشارة المخرج لنظام (first order) ومدخل (step):

1. التخلص من feedback loop.

2. تحديد قيمة المدخل وقيمة gain، ثم حساب القيمة النهائية لإشارة المخرج (O_{ss}).

$$O_{ss} = gain \cdot input$$

3. تحديد قيمة ثابت الزمن T، ثم حساب قيمة زمن الاستقرار (t_{ss}).

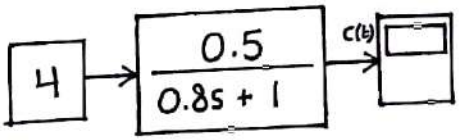
$$t_{ss} = 5T$$

4. تحديد النقطة (O_{ss}, t_{ss}).

5. نرسم الفترة اللحظية لإشارة المخرج ← exponential $(0 < t < t_{ss})$

ثم الفترة التي تستقر عندها إشارة المخرج ← constant $(t_{ss} < t < \infty)$

مثال توضيحي ارسم إشارة المخرج للنظام التالي:



• solution:

input = 4

gain = 0.5

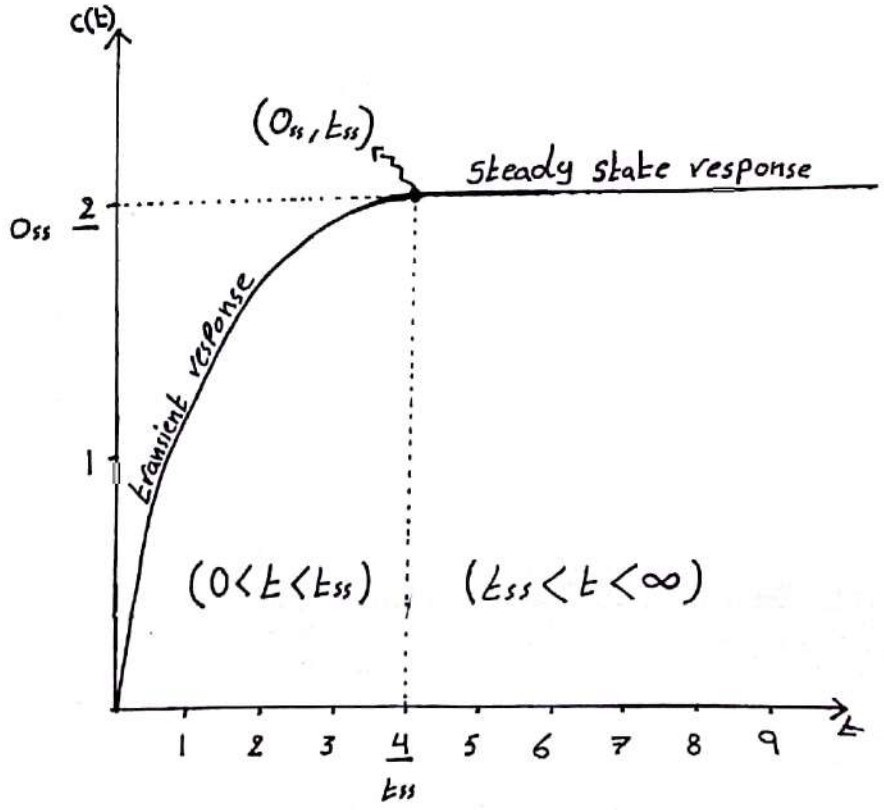
T = 0.8

$O_{ss} = gain \cdot input$

$= (0.5).(4) = 2$

$t_{ss} = 5T$

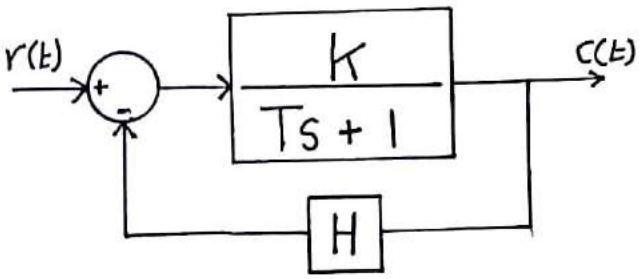
$= 5.(0.8) = 4$



Ex1: Consider the closed loop system below, draw the output signal when:

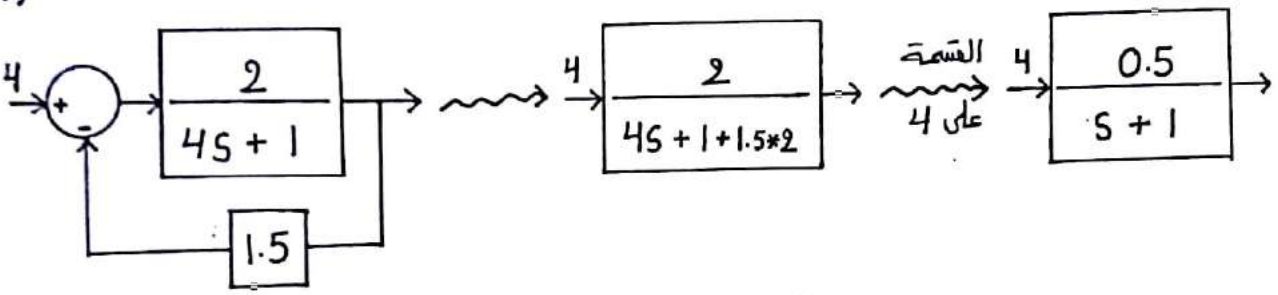
1) $r(t) = 4,\ T = 4,\ H = 1.5,\ K = 2$

2) $r(t) = 1,\ T = 0.9,\ H = 0.6s,\ K = 1$



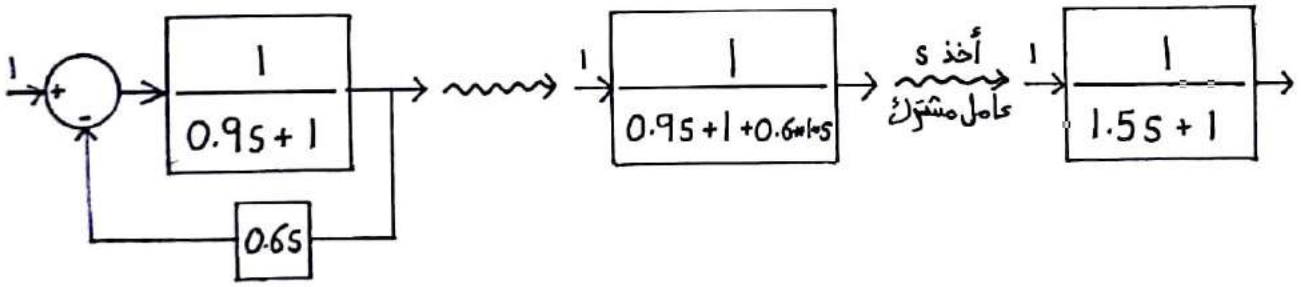
• solution(1):

1)



$O_{ss} = \text{gain} \cdot \text{input} = (0.5) \cdot (4) = 2$ $\qquad t_{ss} = 5T = 5 \cdot (1) = 5$

2)



$O_{ss} = \text{gain} \cdot \text{input} = (1) \cdot (1) = 1$ $\qquad t_{ss} = 5T = 5 \cdot (1.5) = 7.5$

* ملاحظة: نستخدم في الرسم ($t_{ss} = 5T$) لإنه عند هذه اللحظة الزمنية قيمة إشارة المخرج تصل إلى مايقارب 99.3٪ من القيمة النهائية لإشارة المخرج (O_{ss}).

• solution(2):

* Negative feedback *

1) Feedback = 1.5 ⟶ Fixed feedback

$$K_{new} = \frac{K_{old}}{1 + a K_{old}} = \frac{2}{1 + 1.5*2} = \frac{2}{4} = 0.5$$

$$T_{new} = \frac{T_{old}}{1 + a K_{old}} = \frac{4}{1 + 1.5*2} = \frac{4}{4} = 1$$

O_{ss} = gain . input = K_{new} . input = (0.5) . (4) = 2

t_{ss} = 5 . T_{new} = (5) . (1) = 5

2) Feedback = 0.6 s ⟶ flexible feedback

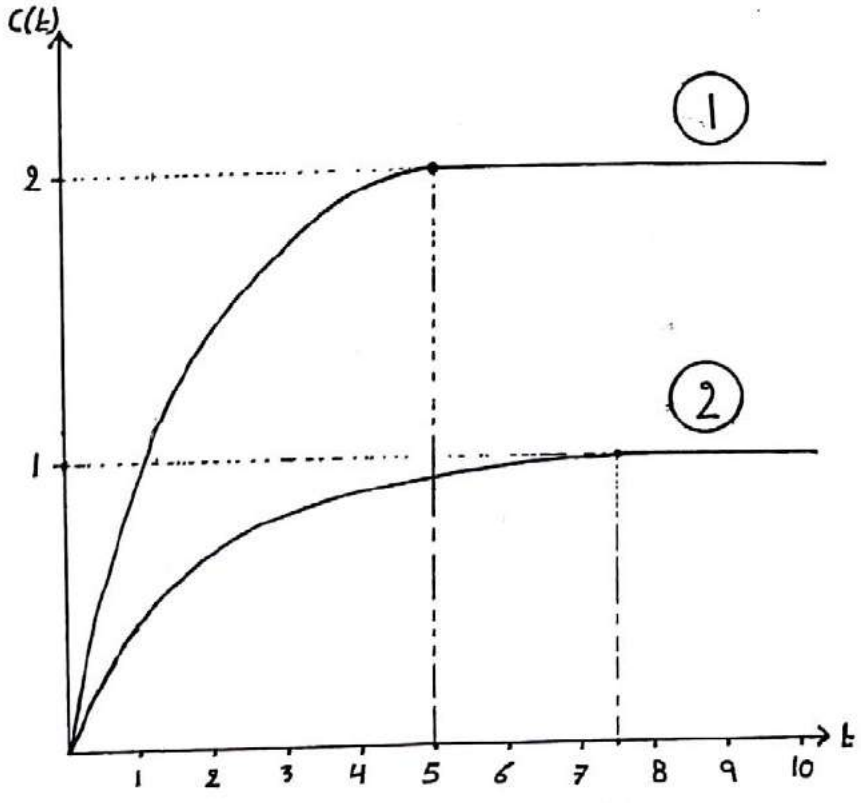
$K_{new} = K_{old} = 1$

$T_{new} = T_{old} + bK_{old} = 0.9 + (0.6) . (1) = 1.5$

O_{ss} = gain . input = (1) . (1) = 1

t_{ss} = 5 . T_{new} = (5) . (1.5) = 7.5

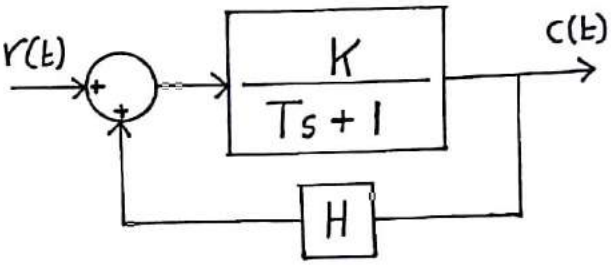
• output:



Ex₂: Consider the closed loop system below, draw the output signal when:

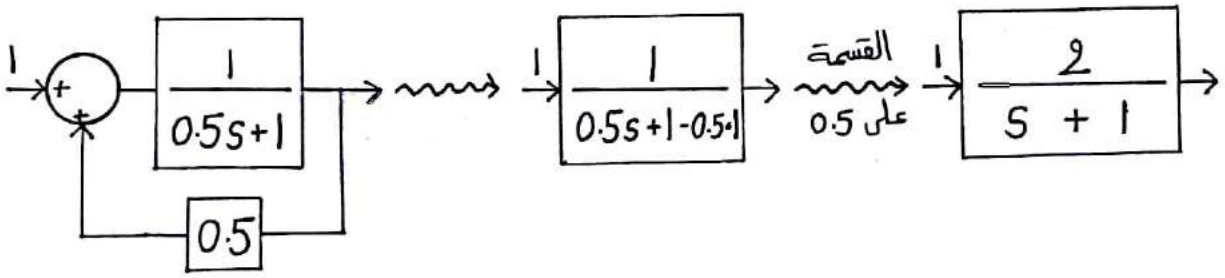
1) $r(t)=1$, $T=0.5$, $H=0.5$, $K=1$

2) $r(t)=1$, $T=2.5$, $H=1$, $K=1$



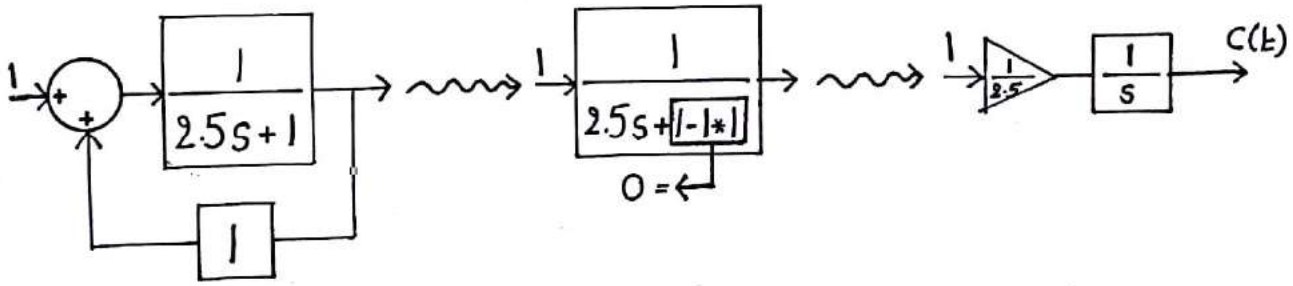
• solution(1):

1)



O_{ss} = gain.input = (2).(1) = 2 $\qquad$ $t_{ss} = 5T = 5.(1) = 5$

2)



$C(t) = \int \frac{1}{2.5} * 1 \cdot dt = 0.4 \int 1 \cdot dt = 0.4t$

$C(t) = 0.4t$ [linear function] , Slope = 0.4

$C(0) = 0$

$C(5) = 2$

$C(10) = 4$

• Solution (2):

Positive Feedback

1) Feedback = 0.5 ⟶ Fixed Feedback ($1-ak>0$ → system is stable)

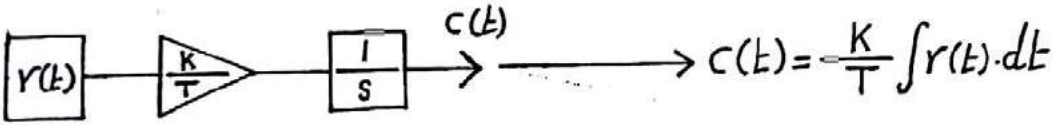
$$K_{new} = \frac{K_{old}}{1-ak} = \frac{1}{1-0.5*1} = \frac{1}{0.5} = 2$$

$$T_{new} = \frac{T_{old}}{1-ak} = \frac{0.5}{1-0.5*1} = \frac{0.5}{0.5} = 1$$

$$O_{ss} = \text{gain.input} = K_{new}.\text{input} = (2).(1) = 2$$

$$t_{ss} = 5\,T_{new} = 5.(1) = 5$$

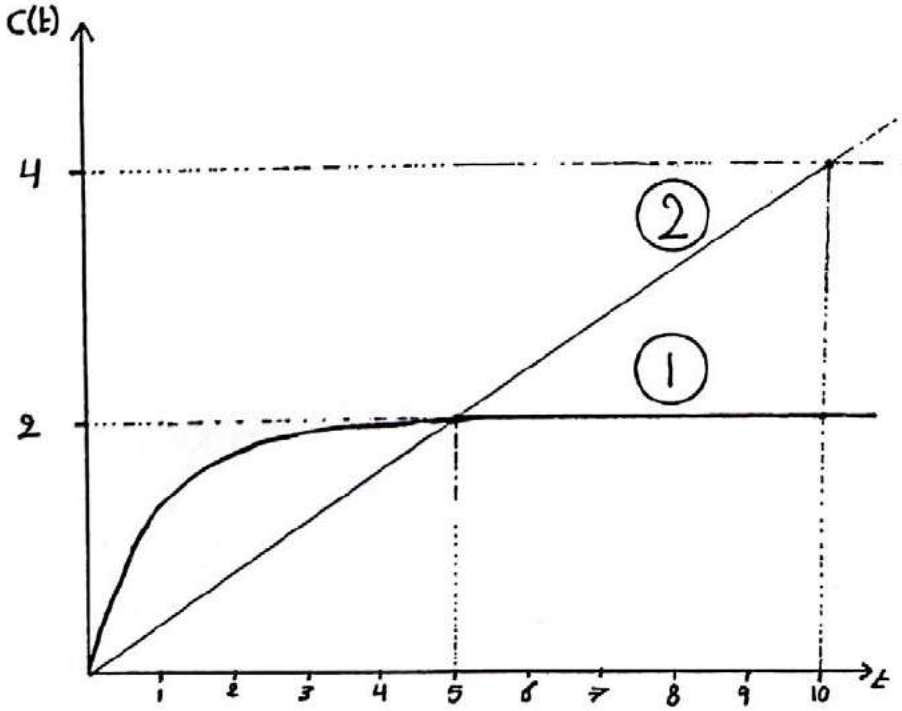
2) Feedback = 1 ⟶ Fixed Feedback ($1-ak=0$ → First order system ⟿ integrator)



$$C(t) = \frac{K}{T}\int r(t).dt$$

$$C(t) = \frac{1}{2.5}\int 1.dt \longrightarrow C(t) = 0.4t$$

• Output:



Experiment (4)
time response of first order system

A. transient response (dynamic response)

1. Step input

a. varying input

- Output steady state value (O_{ss}) = gain . input

$O_{ss} \propto$ input

input (increase) $\longrightarrow$ O_{ss} (increase)

- Steady state error (e_{ss}) = input - O_{ss} = input - gain . input = input (1 - gain)

$e_{ss} \propto$ input

input (increase) $\longrightarrow$ e_{ss} (increase)

b. varying time constant (T)

- settling time (t_{ss}) = 5 T

$t_{ss} \propto T$

T (increase) $\longrightarrow$ t_{ss} (increase)

2. Ramp input

- varying time constant (T)

settling time (t_{ss}) = 5T

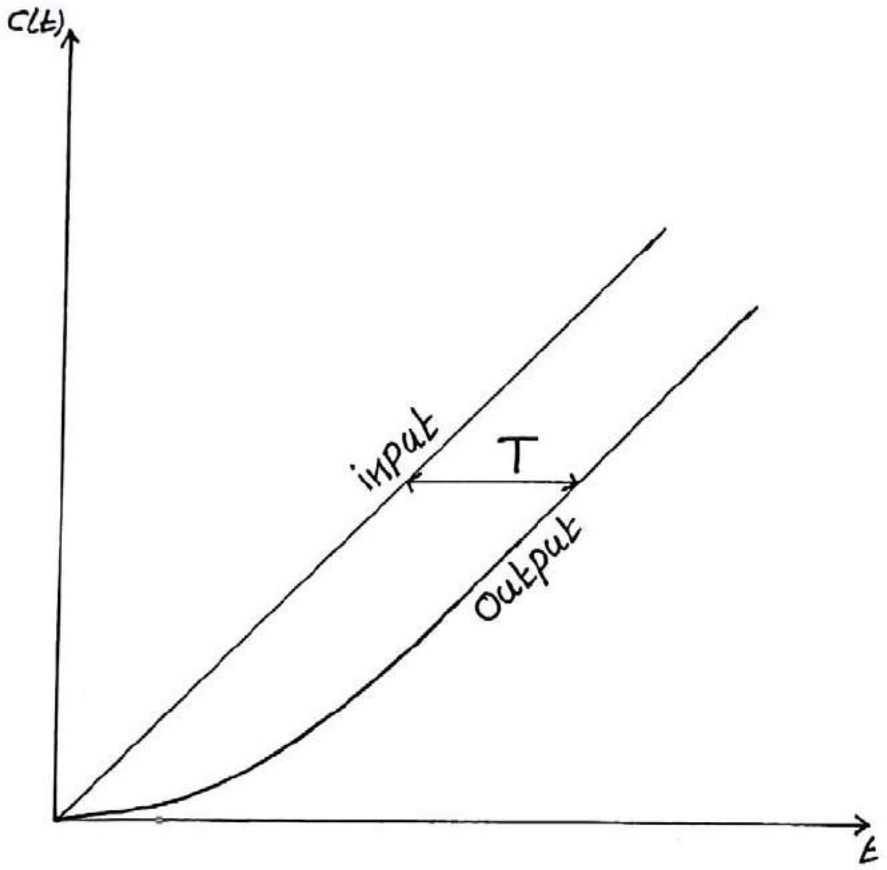
$t_{ss} \propto T$

T (increase) $\longrightarrow$ t_{ss} (increase)

Steady state error (e_{ss}) = T

$e_{ss} \propto T$

T (increase) ⟶ e_{ss} (increase)

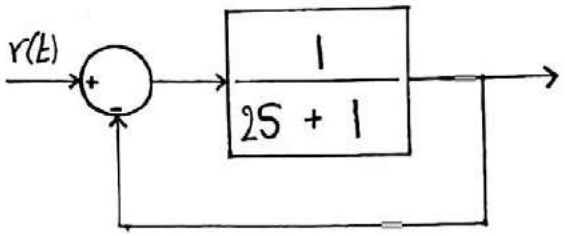


B. Steady state response (static response)

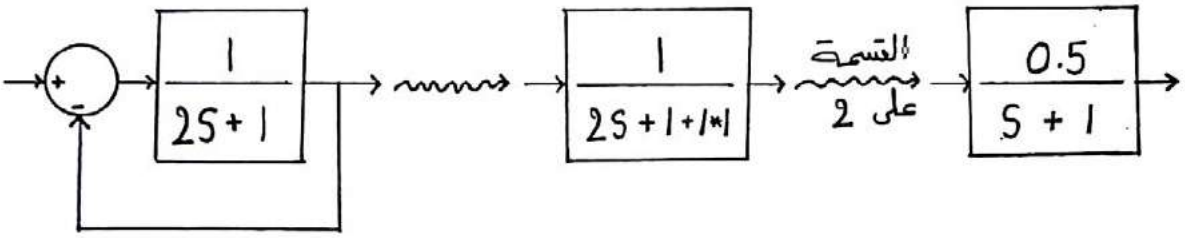
- it can be calculated from the differential equation by setting all derivatives to zero, or from the transfer function by setting s=0.

Ex_1: Consider the system below, calculate the steady state error (e_{ss}) when:

1) $r(t)=1$ 2) $r(t)=2$ 3) $r(t)=4$



• solution:



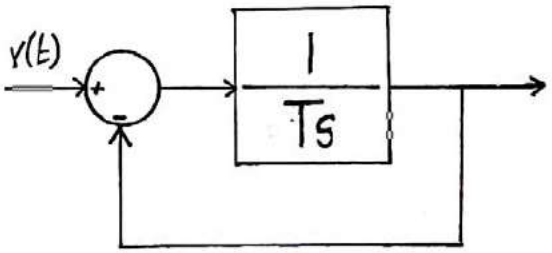
1) e_{ss} = input(1 - gain)
= 1(1-0.5)
= 0.5

2) e_{ss} = input(1-gain)
= 2(1-0.5)
= 1

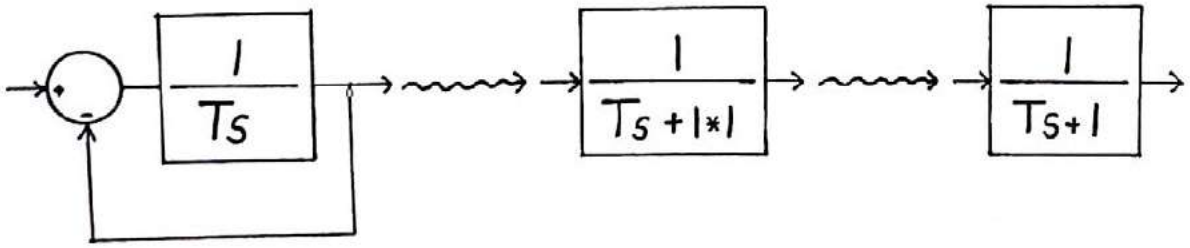
3) e_{ss} = input(1-gain)
= 4(1-0.5)
= 2

Ex_2: Consider the system below, assume $r(t)=t$, calculate the steady state error (e_{ss}) when:

1) $T=1$ 2) $T=0.5$ 3) $T=3$



• Solution:



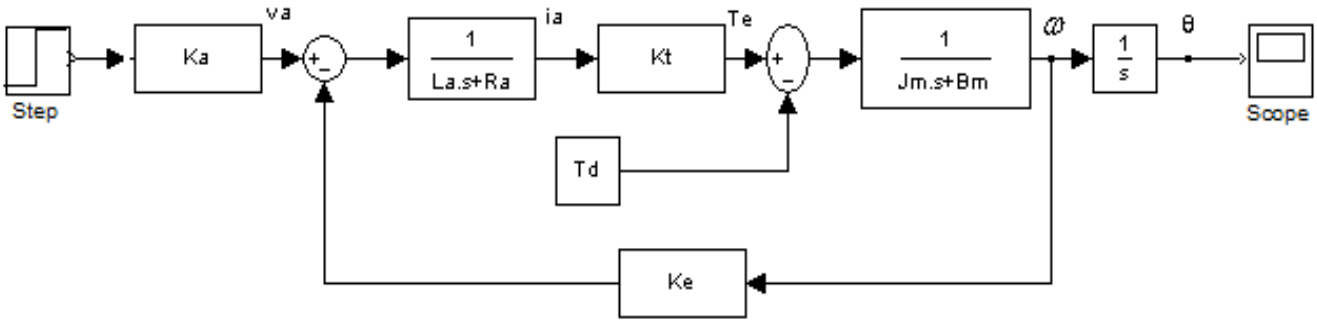
1) $e_{ss} = T$
= 1

2) $e_{ss} = T$
= 0.5

3) $e_{ss} = T$
= 3

EXPeriment (5)
DC servo Motor

Block Digram of DC servo motor



Vi : Input voltage
ia : Armature current
Te : Motor Torque
w : Angular Speed
Ra : Armature Resistance
La : Armature Inducatance

$\frac{1}{La.s + Ra}$: Motor

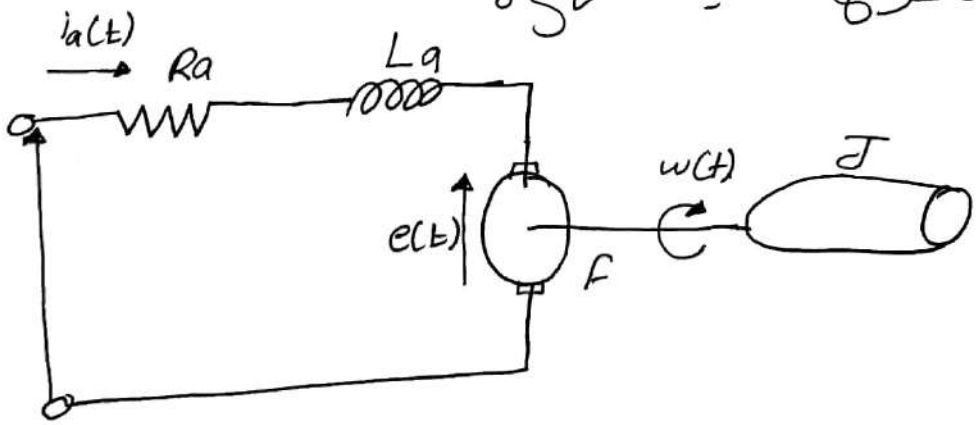
Kt : Motor Torque constant

Va : Armature voltage
Ka : Amplifier
Td : Load Torque
θ : Angular position
Jm : Net Inertia
Bm : Net Damping

$\frac{1}{Jm.s + Bm}$: Gear

Ke : Motor Back emf constant

* شرح لغاية العمل:



بتطبيق قانون كيرتشوف للتيار نحصل على المعادلة التالية :-

$$-Va(t) + ia(t).Ra + La\frac{d[ia(t)]}{dt} + e(t) = 0$$

$$Va(t) = e(t) + ia(t).Ra + La\frac{d[ia(t)]}{dt}$$

لغايات تسهيل الفهم سندرس حالة Steady State :

$$t = \infty \longrightarrow \frac{d[ia(t)]}{dt} = 0$$

$t = \infty \longrightarrow$ Va(t), e(t), ia(t) = قيم ثابتة

تصبح المعادلة :

$$Va = Ea + IaRa$$

$$Ea = Va - IaRa$$

$Ea = Ke.w$ ..... من معادلة (5) في الميكانيك

$$Ke.w = Va - IaRa$$

تم تعديل هذه المعادلة، ولكن تأكدوا منها أيضًا
التعديل بواسطة ليث عمرو وشكرًا لحلمي نزّال ❤

$$* \omega = \frac{V_a}{K_e} - \frac{i_a * R}{K_e} \longrightarrow (Angular\ speed\ equation)$$

$Te = Kt.Ia$ حالة Steady State من معادلة رقم (3) في الميكانيك

$$Jm\frac{dw(t)}{dt} = Te - Td - Bm.w(t)$$ من معادلة رقم (4) في الميكانيك

At $(t=\infty) \longrightarrow \frac{\partial w(t)}{\partial t} = 0$ and $Bm \cdot w(t) = 0$

(Gear box Torque + Friction Torque ≈ 0)

$Td = Te = Kt \cdot Ia$ (Load Torque equation)

A) Transient Response of DC Motor

من معادلة السرعة (Angular Speed equation) ومعادلة عزم الحمل (Load Torque equation) نستنتج أنه العلاقة بين التيار (Ia) في العزم (Td)، والعلاقة بين التيار (Ia) والسرعة (w) عكسية

$$Td = Kt \cdot Ia \longrightarrow Td \propto Ia$$

$$\omega = \frac{V_a}{K_e} - \frac{i_a * R}{K_e} \longrightarrow w \frac{1}{\propto} Ia$$

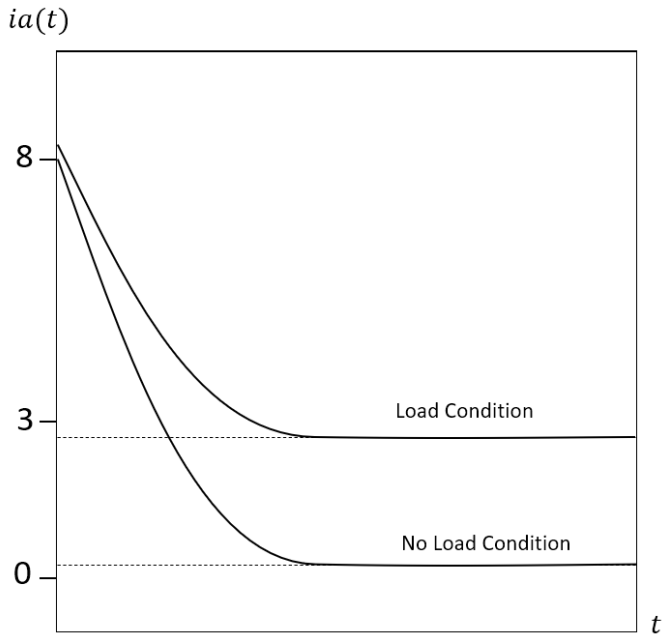
$* \; Td \frac{1}{\propto} w$

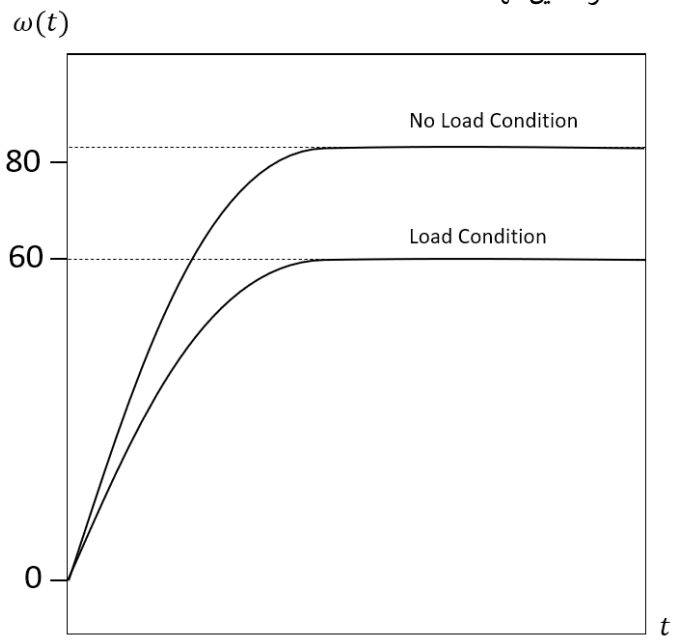
Td (increase) $\longrightarrow$ Ia (increase) $\longrightarrow$ w (decrease)

Td (decrease) $\longrightarrow$ Ia (decrease) $\longrightarrow$ w (increase)

Td = 0 (no load condition) $\longrightarrow$ Ia (minimum $Ia = \frac{Bm \cdot w}{ke} \approx 0$) $\longrightarrow$ w (max)

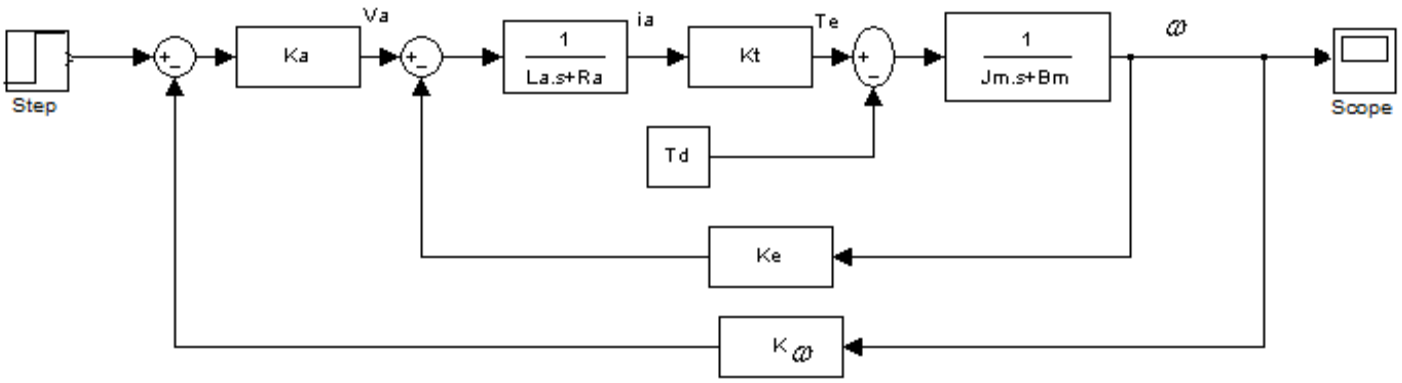
*الرسمتين مهمات!





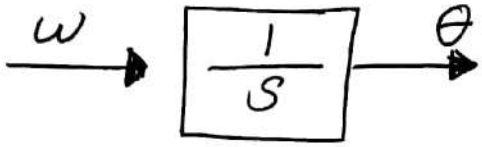
- No load condition (Td= 0 N.m)
- Load condition (Td= 1.147N.m)

B) Transient Response of Speed closed Loop System



عند إضافة Speed Feedback فإن استجابة النظام تصبح أسرع (tss decrease)

C) Transient Response of Position Closed Loop system.



$\int$ Angluar speed = Angular position $\longrightarrow \int w(t).dt = \theta(t)$.

$w \propto \theta$ and $Td \frac{1}{\propto} w \longrightarrow Td \frac{1}{\propto} \theta$

Td (increase) $\longrightarrow$ Ia (increase) $\longrightarrow$ w (decrease) $\longrightarrow$ θ (decrease)

Td (decrease) $\longrightarrow$ Ia (decrease) $\longrightarrow$ w (increase) $\longrightarrow$ θ (increase)

Ex1: Explain what happens in the following conditions:

a) Load torque changed from 1.147 N.m to 0.5 N.m

b) Increasing the load torque.

Solution:

a) Td decrease from 1.147 N.m to 05 N.m, when the load torque decrease the armature current decreases on the other hand the angular speed increases.

b) when the load torque is increased the armature current increase on the other hand the angular speed decreases.

H.W1 : Sketch i(t) and w(t) curves in case of position closed loop system for load condition (Td = 1.147) and no load condition (Td=0)

مساعدة : أنشئ المودل التالي على الماتلاب باستعمال القيم الموجودة في الجدول (1) في الملف الوورد

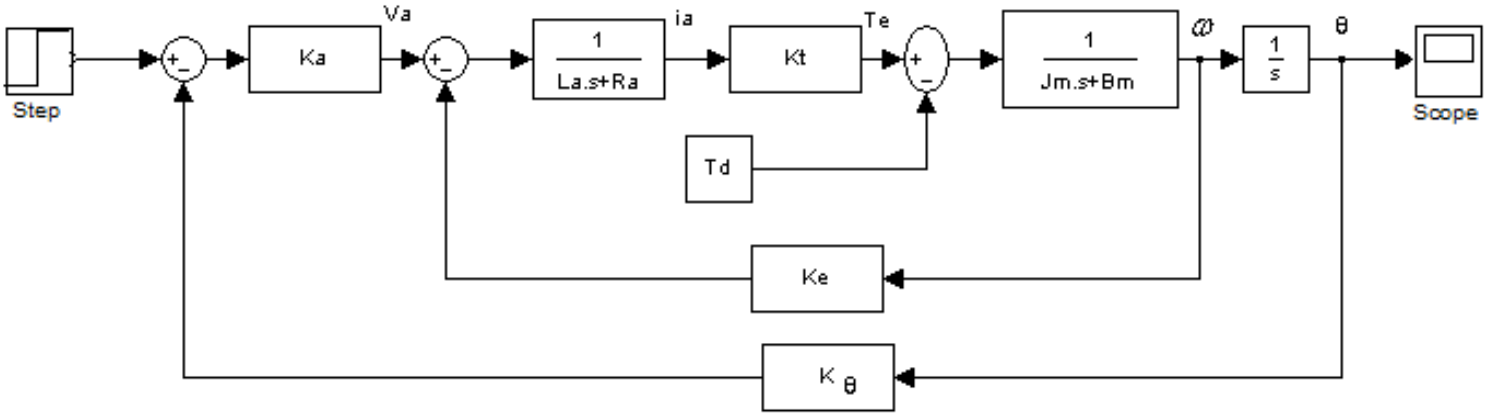


TABLE1: Motor rated output and nominal parameters

Ka = 40 V/V	Ra = 4.7(Ω)
ke = 0.45 V/(rad/s)	La = 0.011 H
Td = 1.147 Nm	Jm = 0.0098 kg m²
kt = 0.45 Nm/A	Bm = 0.0015 Nm/(rad/s)
kw = 0.1	kθ = 0.1

Mid Exam
(Experiment 1, 2, 3, 4, 5)

Q.1: Choose the correct answer: [9 Marks]

1. In transfer function $\frac{K}{Ts+A}$, the constants K, T, A respectively represent:

a. time constant, gain, fixed feedback.
b. gain, time constant, fixed feedback.
c. time constant, gain, flexible feedback.
d. gain, time constant, flexible feedback.

2. We represent flexible feedback in MATLAB Simulink as:

a. constant.
b. transfer function.
c. gain.
d. gain + derivative.

3. If we decrease load:

a. the speed increase and current decrease.
b. the speed decrease and current increase.
c. the speed and current increase.
d. the speed and current decrease.

4. If we increase load:

a. the position increase and current decrease.
b. the position and current decrease.
c. the position decrease and current increase.
d. the speed and current decrease.

5. If we integrated speed ($\boldsymbol{\omega}$) we get:

a. velocity.
b. acceleration.
c. inertia
d. position.

6. The transfer function $\frac{1}{Jm.s+Bm}$ represent:

a. amplifier.
b. electrical motor.
c. gear.
d. load.

Q.2: [7 Mark]

a. To represent the transfer function $\frac{9.s+28}{2.s^3}$ in MATLAB Simulink: [1 Mark]

denominator matrix [ ]
numerator matrix [ ]

b. Draw the simulation diagram for the following: [6 Marks]

1. $y(t) = 5 - t^2 + 0.1 \int -y(t).dt$

2. $-2y''(t) + 4z(t).y(t) - 6y'(t) = -2$

$z''(t) + 2z'(t).y'(t) - 0.5z(t) = 2$

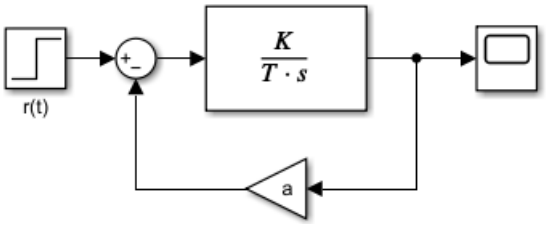
Q.3: For the system below, plot the output signal c(t) when: [6 Marks]

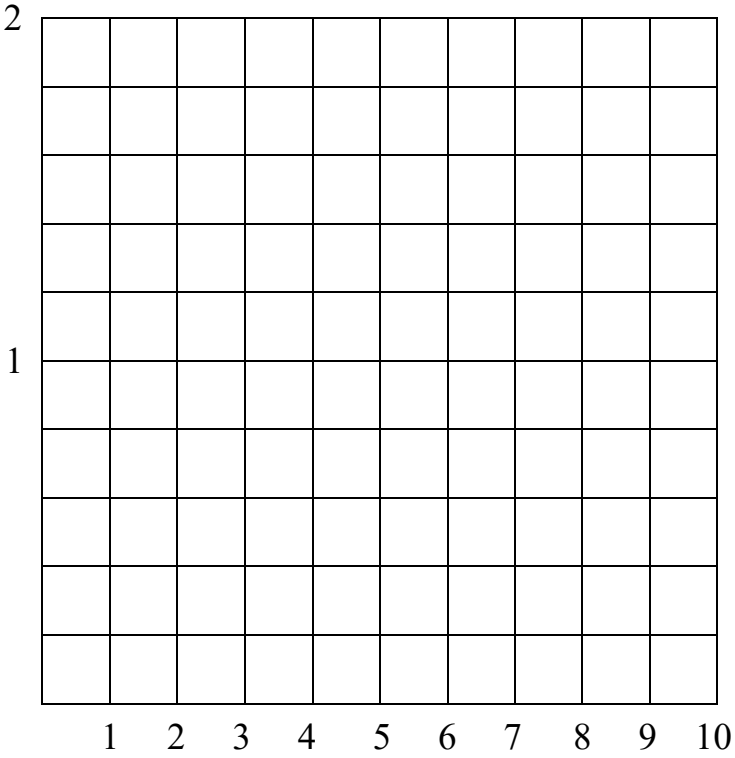
a. r(t) = 1.2 , T = 1 , a = 2 , K = 1

b. r(t) = 1.4 , T = 1 , a = 1 , K = 1

c. r(t) = 1.0 , T = 5 , a = 0 , K = 1

• System:





Q.4: [8 Marks]

a. For the Equation $y''' + 2y'' - 4y' + 3y = u(t)$, find the matrices [A, B, C, D] and enter their values as in MATLAB Simulink [6 Marks]

A = [ ]

B = [ ]

C = [ ]

D = [ ]

b. Find the transfer function for the equation below: [2 Marks]

$-2y''' + 4y'' - 6y' + 8y = -4r(t)$

Mid Exam Answers

Q.1:

1. B	2. D	3. A
4. C	5. D	6. C

Q.2:

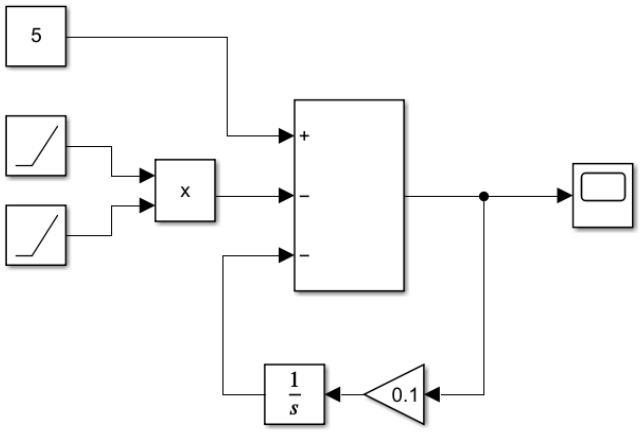
a. denominator matrix [2 0 0 0]
numerator matrix [9 28]

b.

1.

$$y(t) = 5 - t^2 + 0.1\int -y(t).dt$$

$$y(t) = 5 - t^2 - 0.1\int y(t).dt$$



2.

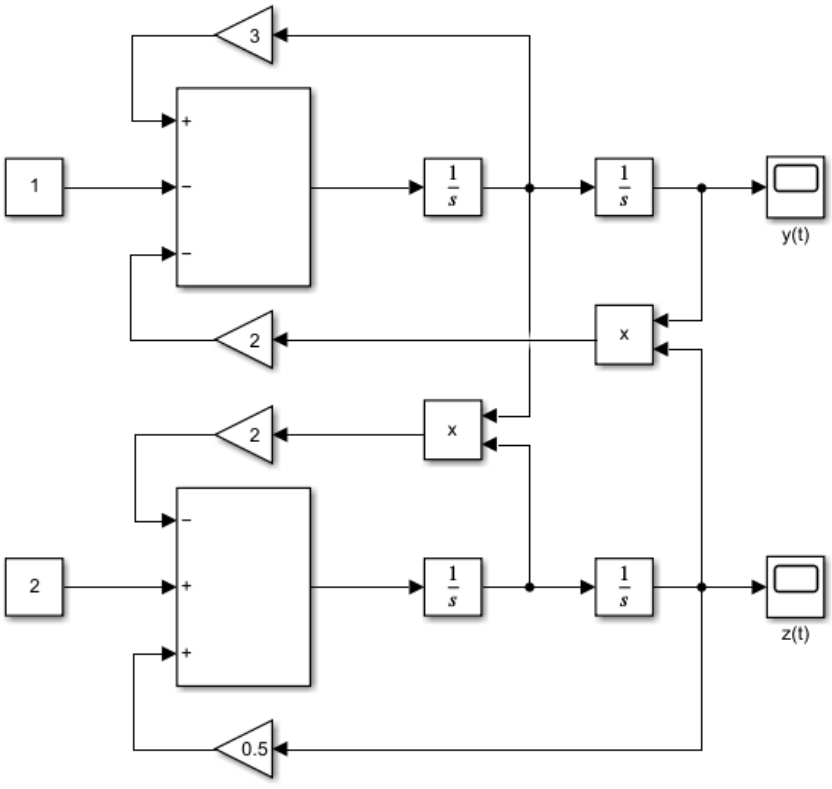
$$-2y''(t) + 4z(t).y(t) - 6y'(t) = -2$$

$$y''(t) + 2z(t).y(t) + 3y'(t) = 1$$

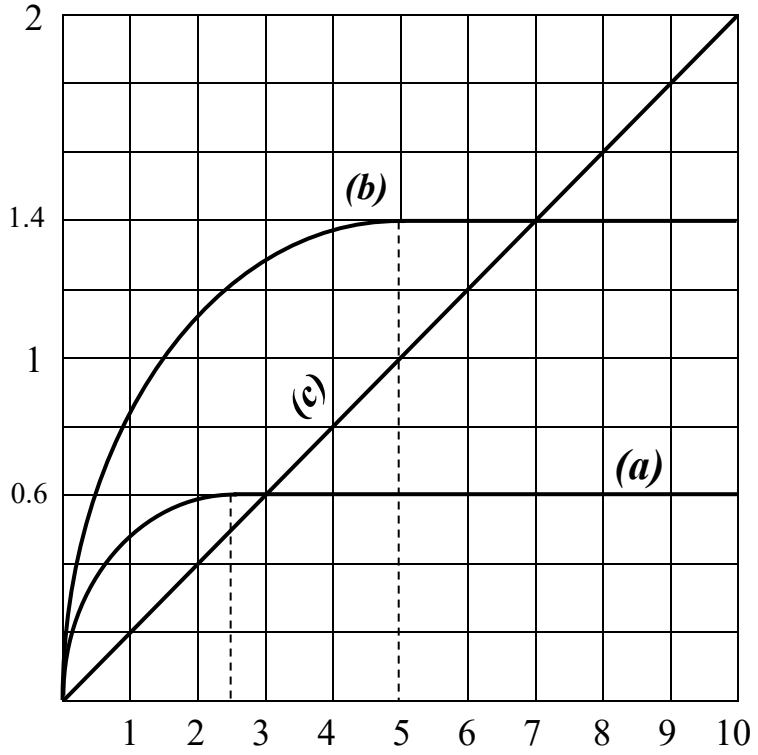
$$y''(t) = 1 - 2z(t).y(t) - 3y'(t)$$

$$z''(t) + 2z'(t).y'(t) - 0.5z(t) = 2$$

$$z''(t) = 2 - 2z'(t).y'(t) + 0.5z(t)$$



Q.3:



Q.4:

a.

A = [0 1 0 ; 0 0 1 ; -3 4 2]

B = [0 ; 0 ; 1]

C = [1 0 0]

D = [0]

b.

$-2y''' + 4y'' - 6y' + 8y = -4r(t)$

$y''' - 2y'' + 3y' - 4y = 2r(t)$

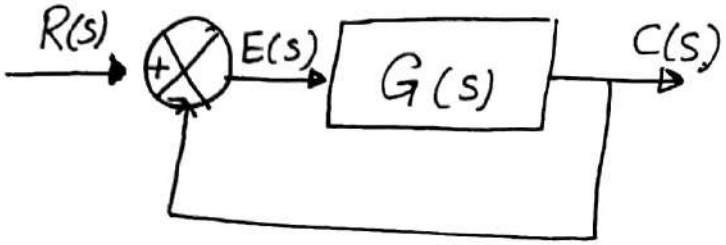
$s^3.Y(s) - 2s^2.Y(s) + 3s.Y(s) - 4Y(s) = 2R(s)$

$Y(s)\,[\,s^3 - 2s^2 + 3s - 4\,] = 2R(s)$

$$\frac{Y(s)}{R(s)} = \frac{2}{s^3 - 2s^2 + 3s - 4}$$

Experiment (6)

Time Response of Second order system



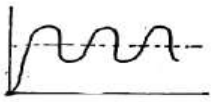
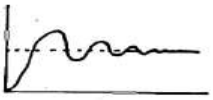
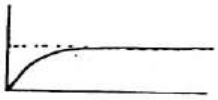
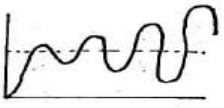
$$G(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)} \xrightarrow[\text{elimination}]{\text{feedback}} G(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n) + 1 * \omega_n^2}$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega + \omega_n^2}$$

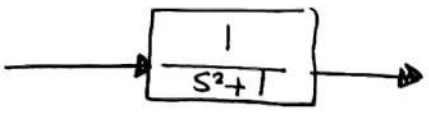
ω_n ⟶ natural frequecy

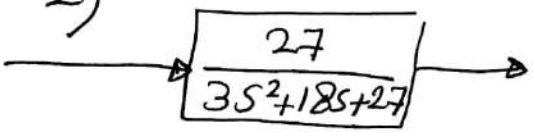
ζ ⟶ damping ratio (zeta)

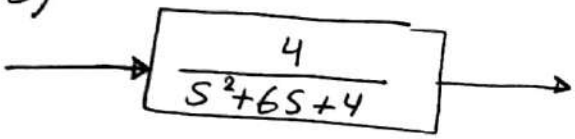
A) The effect of the dampling ratio variation on the system response.

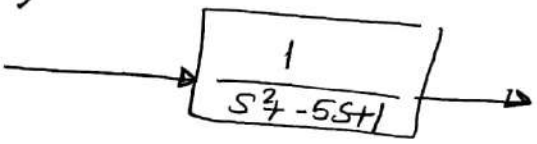
damping Ratio	response type	output signal
1) $\zeta = 0$	undamped	
2) $0 < \zeta < 1$	under damped	
3) $\zeta = 1$	critically damped	
4) $\zeta > 1$	over damped	
5) $\zeta < 0$	unstable	

Ex: determine the response type of the following systems:

1) 

2) 

3) 

4) 

* Solution:

1) $\omega_n^2 = 1 \longrightarrow \omega_n = 1$

$2\xi\omega_n = 0 \longrightarrow \xi = 0$

Type of response ⟶ Undamped ($\xi = 0$)

2) $\frac{27}{3S^2+18S+27}$ القسمة على 3 لجعل معامل $S^2 = 1$ $\frac{9}{S^2+6S+9}$

$\omega_n^2 = 9 \longrightarrow \omega_n = 3$

$2\xi\omega_n = 6 \longrightarrow 6\xi = 6 \longrightarrow \xi = 1$

Type of response ⟶ Critically Damped ($\xi = 1$)

3) $\omega_n^2 = 4 \longrightarrow \omega_n = 2$

$2\xi\omega_n = 6 \longrightarrow 4\xi = 6 \longrightarrow \xi = 1.5$

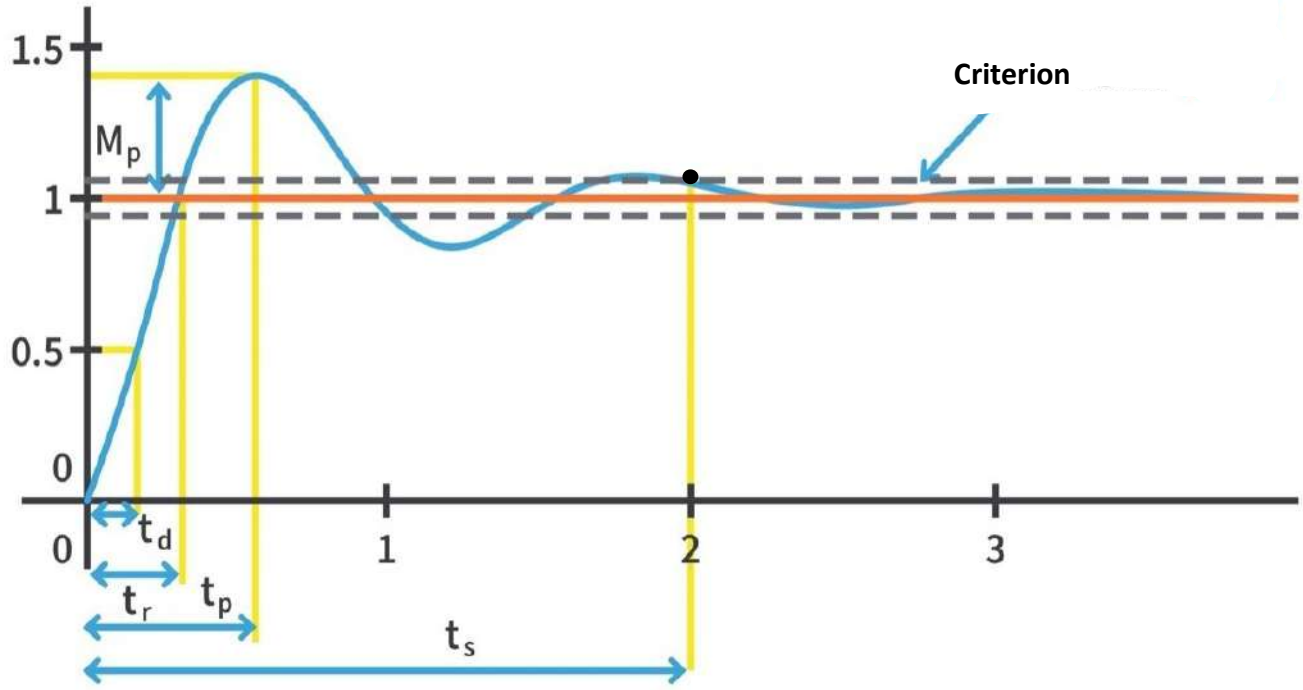
Type of response ⟶ over Damped ($\xi > 1$)

4) $\omega_n^2 = 1 \longrightarrow \omega_n = 1$

$2\xi\omega_n = -5 \longrightarrow 2\xi = -5 \longrightarrow \xi = -2.5$

Type of response ⟶ unstable ($\xi < 0$)

Time Response Specification :



* t_d (Delay time) : هو الوقت الذي تستغرقه الموجة لتصل إلى نصف القيمة النهائية (Steady state value)

* t_r (Rise time): هو الوقت الذي تستغرقه الموجة لتقطع خط (Steady State) لأول مرة

* t_p (peak time): هو الوقت الذي تستغرقه الموجة للوصول للقيمة العليا (Peak Value)

* t_{ss} (setting time/ steady state time):
هو الوقت الذي تستغرقه الموجة لكي تستقر عند القيمة النهائية (Steady State Value)

* t_{ss} (n% criterion) :
هو الوقت اللازم للموجة للوصول إلى أول انحطاف قبل انحصارها ضمن حدود ± n% من القيمة النهائية

* Mp (Maximum peak overshoot):
هو الفرق بين قيمة الموجة عند القمة و القيمة النهائية للموجة

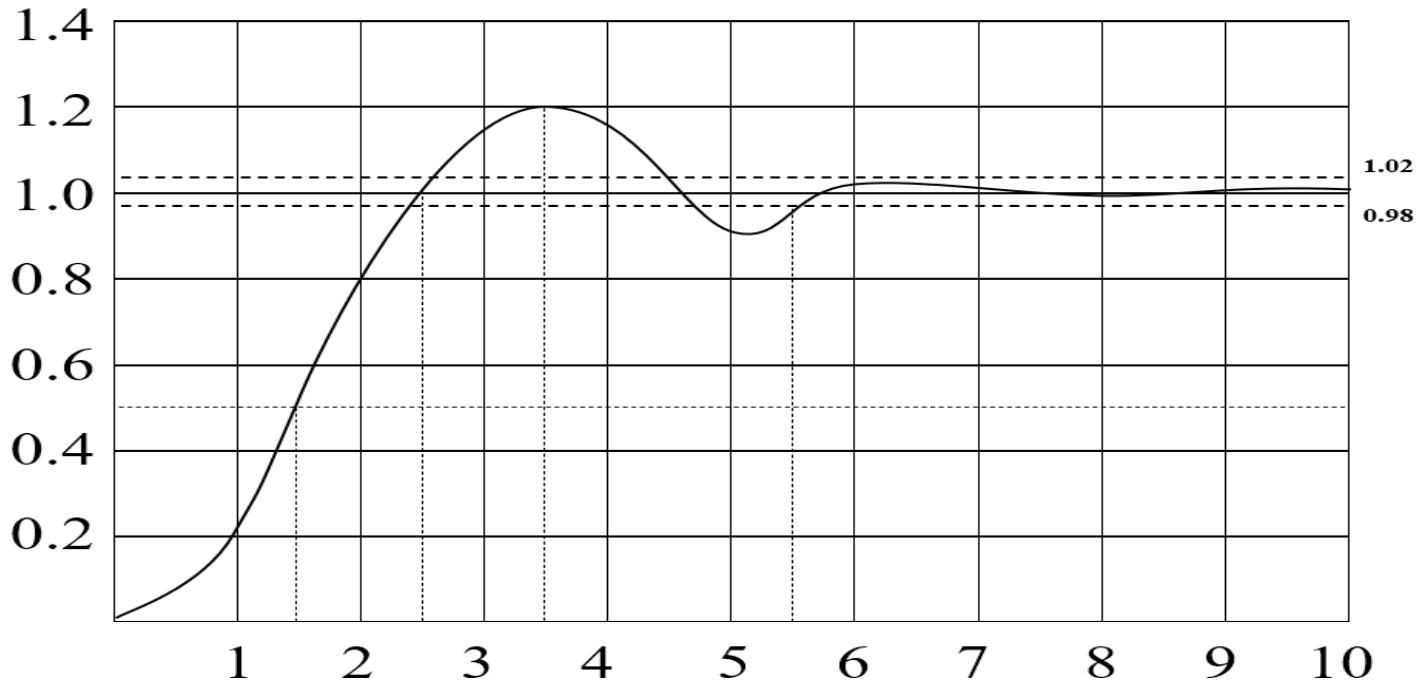
Mp = peak value - Steady State Value

* Mp%: هي نسبة الزيادة المئوية للموجة عند القيمة النهائية

$$Mp\% = \frac{\text{Peak Value} - \text{Steady State Value}}{\text{Steady State Value}} \% = \frac{Mp}{\text{steady State Value}} \%$$

* إيجاد (Time response specifications) عن طريق الرسمة:

Ex2: Depending on the graph Below, find Td, Tr, tp, tss (2% criterion), Mp.



• Solution:

1) Td → زمن وصول الموجة إلى نصف القيمة النهائية

$t_d = 1.5s$

2) t_r ⟵ زمن قطع الموجة خط القيمة النهائية لأول مرة

$t_r = 2.5s$

3) t_p ⟵ زمن وصول الموجة إلى أعلى قيمة لها (القمة)

$t_p = 3.5s$

4) $t_{ss(2\%)}$ ⟵ زمن وصول الموجة لأول انحصار قبل انحصار الموجة بالحدود ± 2% من القيمة النهائية للموجة

$t_{ss(2\%)} = 5.6s$

5) M_p = Peak value − Steady state value

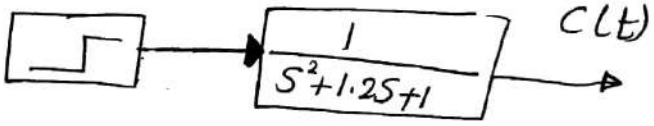
$M_p = 1.2 - 1 = 0.2$

* إيجاد (Time respone specifiation) عن طريق (Transfer Function) باستخدام المعادلات التالية:

* $\theta = \cos^{-1}(\xi)$ * [θ is in radation]
* $w_d = w_n\sqrt{1-\xi^2}$
* $t_r = \frac{\pi - \theta}{w_d}$
* $t_{ss(2\%)} = 4T = \frac{4}{\xi w_n}$
* $t_p = \frac{\pi}{w_d}$
* $t_{ss(5\%)} = 3T = \frac{3}{\xi w_n}$
* M_p = steady state volue (input) * $e^{\frac{-\pi\xi}{\sqrt{1-\xi^2}}}$

Ex: Calculate the following for the system below assume a unit step = 1:

① t_r 2) t_p 3) t_{ss} (5% criterion) 4) M_p



*Solution:

$\omega_n^2 = 1 \longrightarrow \omega_n = 1$

$2\zeta\omega_n = 1.2 \longrightarrow \zeta = 0.6$

$\theta = \cos^{-1}(\zeta) = \cos^{-1}(0.6) = 0.9273$ radian

$\omega_d = \omega_n\sqrt{1-\zeta^2} = 1*\sqrt{1-0.6^2} = 0.8 \frac{rad}{s}$

1) $t_r = \frac{\pi-\theta}{\omega_d} = \frac{\pi-0.9273}{0.8} = 2.768s$

2) $t_p = \frac{\pi}{\omega_d} = \frac{\pi}{0.8} = 3.927s$

3) $t_{ss(5\%)} = 3T = \frac{3}{\zeta\omega_n} = \frac{3}{0.6*1} = 5s$

4) $M_p = input * e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} = 1 * e^{\frac{-\pi(0.6)}{\sqrt{1-0.6^2}}} = 0.09478$

- Hw 1: Find the transfer function that gave the output signal in Ex_2. assumly that $e_{ss}=0$

- Hw2: draw the output signal $c(t)$ in Ex_3

H.w1 Solution:-

$t_p = 3.5$

$t_p = \frac{\pi}{w_d}$

$w_d = \frac{\pi}{t_p} = \frac{\pi}{3.5} = 0.8976$

$t_{ss}(2\%) = 5.6$

$t_{ss}(2\%) = \frac{4}{\zeta w_n}$

$w_n = \frac{4}{\zeta * 5.6}$

$w_d = w_n \sqrt{1-\zeta^2} \rightarrow 0.9 = \frac{4}{\zeta * 5.6} \sqrt{1-\zeta^2}$

$\rightarrow \frac{4}{\zeta * 5.6} \sqrt{1-\zeta^2} - 0.9 = 0$

* كتابة المعادلة ع الآلة الحاسبة (استبدل ζ بـ x)
← الضغط ع Shift ← Calc

$\zeta = 0.6349$

$w_n = \frac{4}{\zeta * 5.6} = \frac{4}{0.6349 * 5.6} = 1.125$

$w_n = 1.125 \rightarrow w_n^2 = 1.266$

$2\zeta w_n = 2 * 0.6349 * 1.266 = 1.608$

$\text{Transfer function} = \frac{w_n^2}{s^2 + 2\zeta w_n s + w_n^2} = \frac{1.266}{s^2 + 1.608s + 1.266}$

* HW2 Solution:

$$\frac{1}{S^2+1.2S+1}$$ T.F

$$\omega_n^2 = 1 \longrightarrow \omega_n = 1$$

$$2\zeta\omega_n = 1.2 \longrightarrow 2\zeta \cdot 1 = 1.2 \longrightarrow \zeta = 0.6$$

$$\theta = \cos^{-1}(0.6) = 0.9273 \text{ rad}$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$= \sqrt{1-0.6^2} = 0.8$$

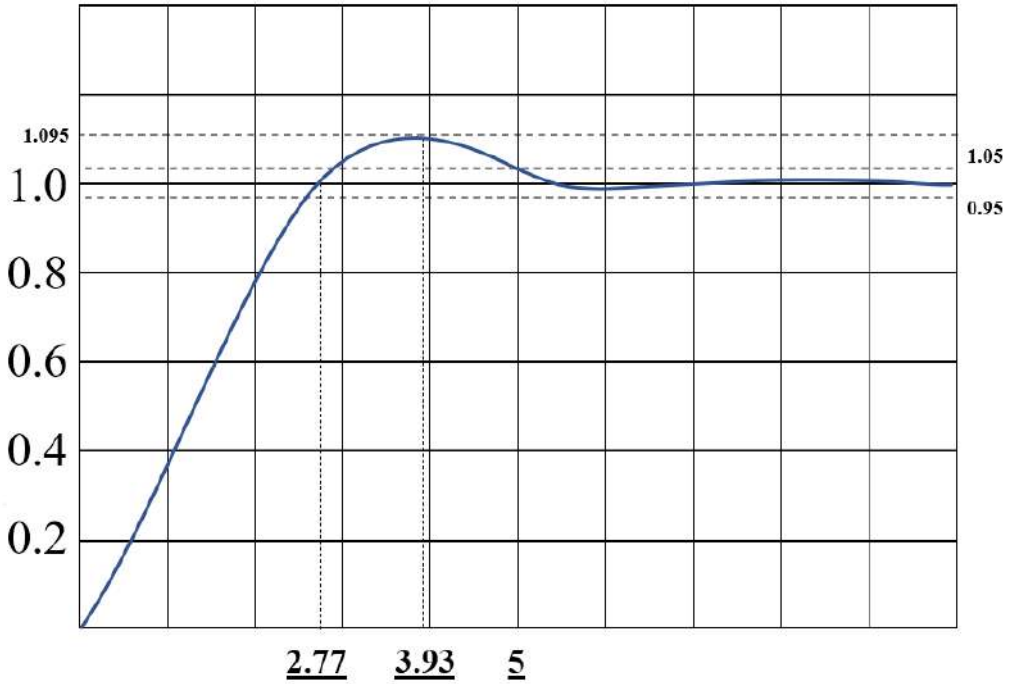
يمكن استخدام أي criterion شريطة توضح ذلك على الرسمة برسم الخطوط ±% criterion ورسم زمن الاستقرار عند نقطة أول نقطة انطلاق قبل الدخول بحدود الـ ± criterion%

* $t_r = \frac{\pi-\theta}{\omega_d} = \frac{\pi-0.9273}{0.8} = 2.775$

$$t_p = \frac{\pi}{\omega_d} = 3.935$$

$$t_{ss(5\%)} = \frac{3}{\zeta\omega_n} = \frac{3}{0.6\,1} = 5s$$

$$M_p = e^{\left[\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right]} = 0.095$$



B) The effect of the frequecy variation on the system response

1) Variation of Natural frequecy (ω_n)

ω_n (increase) ⟶ time (decrease) ⟶ M_p (constant)

ω_n (decrease) ⟶ time (increase) ⟶ M_p (constant)

* time = t_d, t_r, t_p, t_{ss}

* time = $\frac{1}{\text{frequecy}}$

2) Variation of Damping Ratio (ζ)

ζ (increase) ⟶ time (increase) ⟶ M_p (decrease)

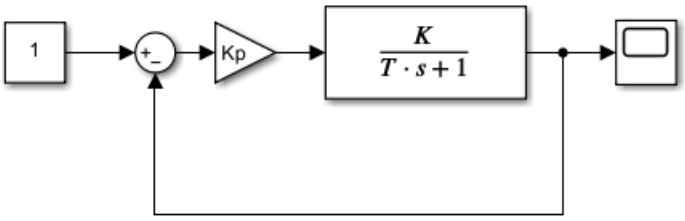
ζ (decrease) ⟶ time (decrease) ⟶ M_p (increase)

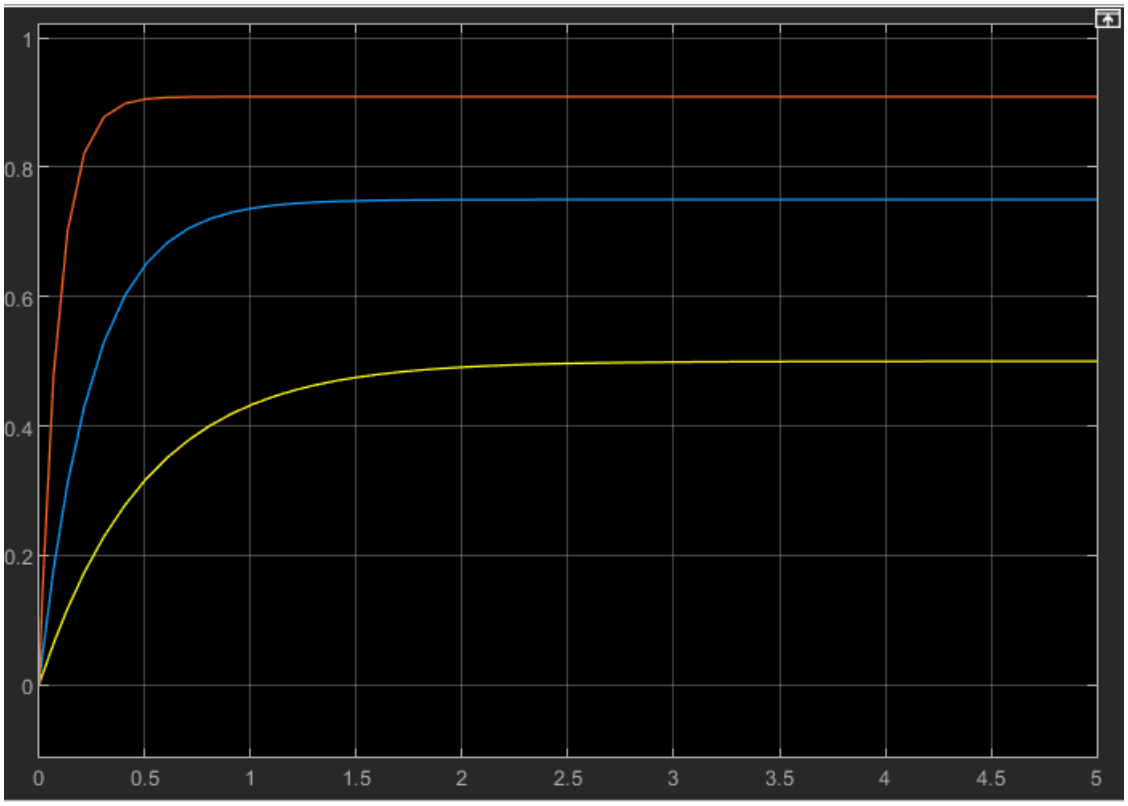
Experiment (7)

Basic Control Action

A. First Order System

1. P Controller





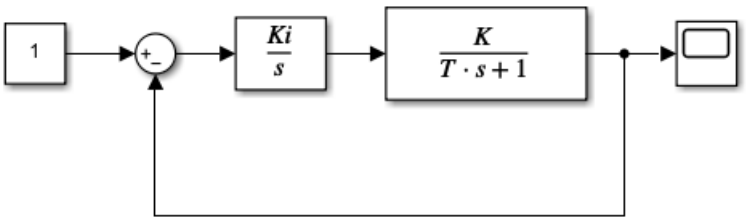
	Kp = 10
	Kp = 3
	Kp = 1

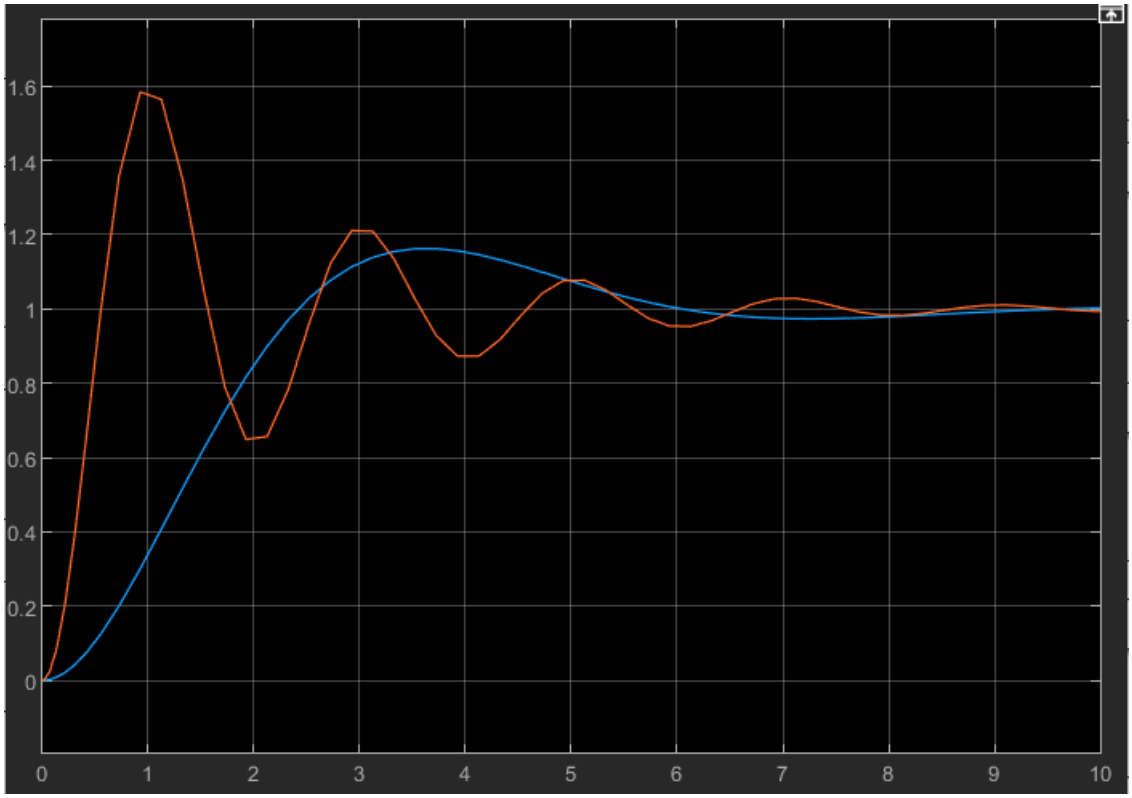
*Notes:

$\boldsymbol{Kp}\ (\boldsymbol{increase}) \rightarrow t_{ss}\ (decrease) \rightarrow e_{ss}(decrease)$

$\boldsymbol{Kp}\ (\boldsymbol{decrease}) \rightarrow t_{ss}\ (increase) \rightarrow e_{ss}(increase)$

2. I Controller





*Notes:

$\boldsymbol{Ki}$ $(\boldsymbol{increase}) \rightarrow Oscillations\ (increase) \rightarrow Mp\ (increase)$
$\rightarrow t_d\ , t_r\ , t_p\ (decrease)$

$\boldsymbol{Ki}$ $(\boldsymbol{decrease}) \rightarrow Oscillations\ (decrease) \rightarrow Mp\ (decrease)$
$\rightarrow t_d\ , t_r\ , t_p\ (increase)$

$*$ $\boldsymbol{Adding\ I\ Controller}$:

$\rightarrow Eliminates\ steady\ state\ error\ [e_{ss} = 0]$

$\rightarrow Makes\ the\ system\ response\ slower$

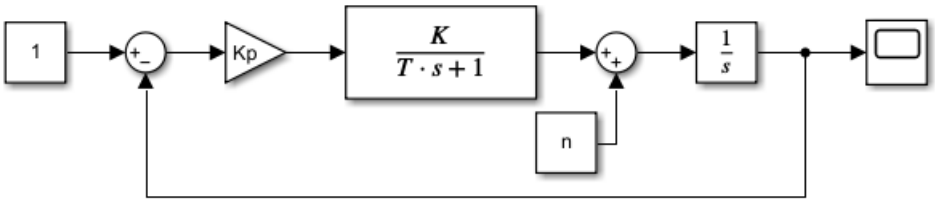
$\rightarrow Changes\ the\ order\ of\ system$:

$[\ First\ Order\ System \rightarrow Second\ Order\ System\ ]$

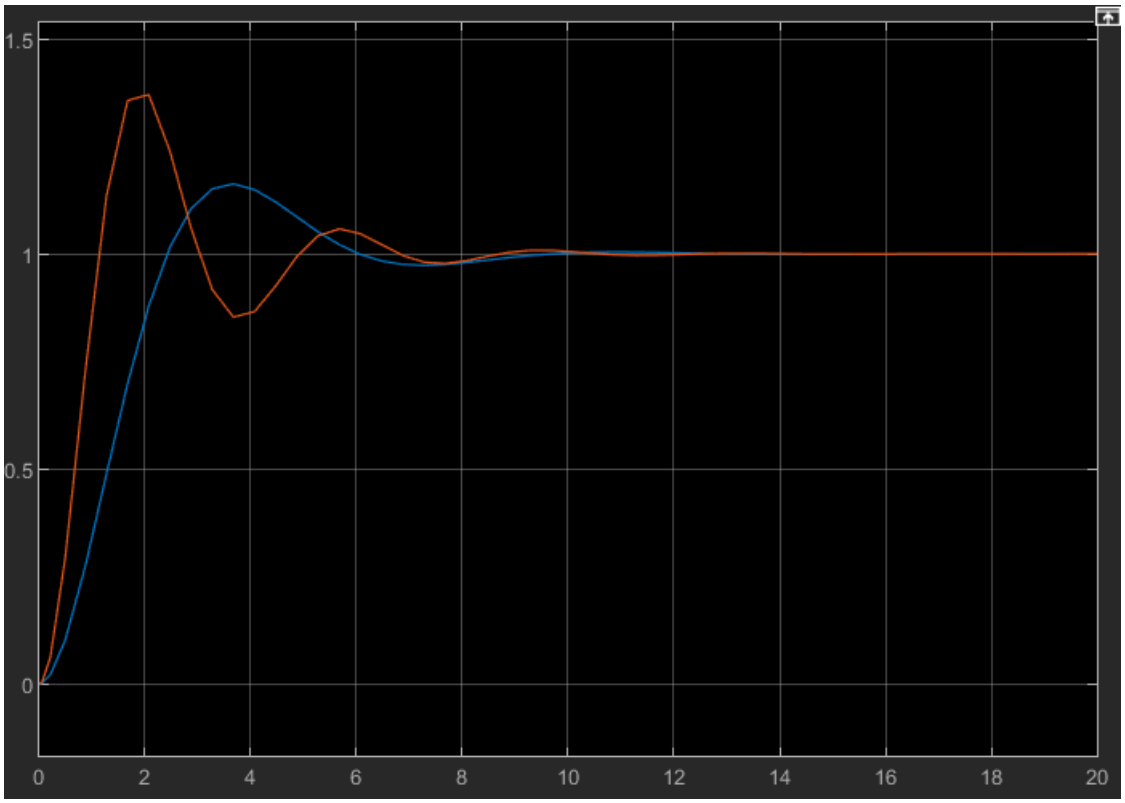
$$\frac{Ki}{s} * \frac{K}{Ts+1} \rightarrow \frac{Ki * K}{s(Ts+1)}$$

B. Second Order System

1. P Controller



• Varying Kp (n = 0):



	Kp = 3
	Kp = 1.2
n = 0 (constant)	

*Notes:

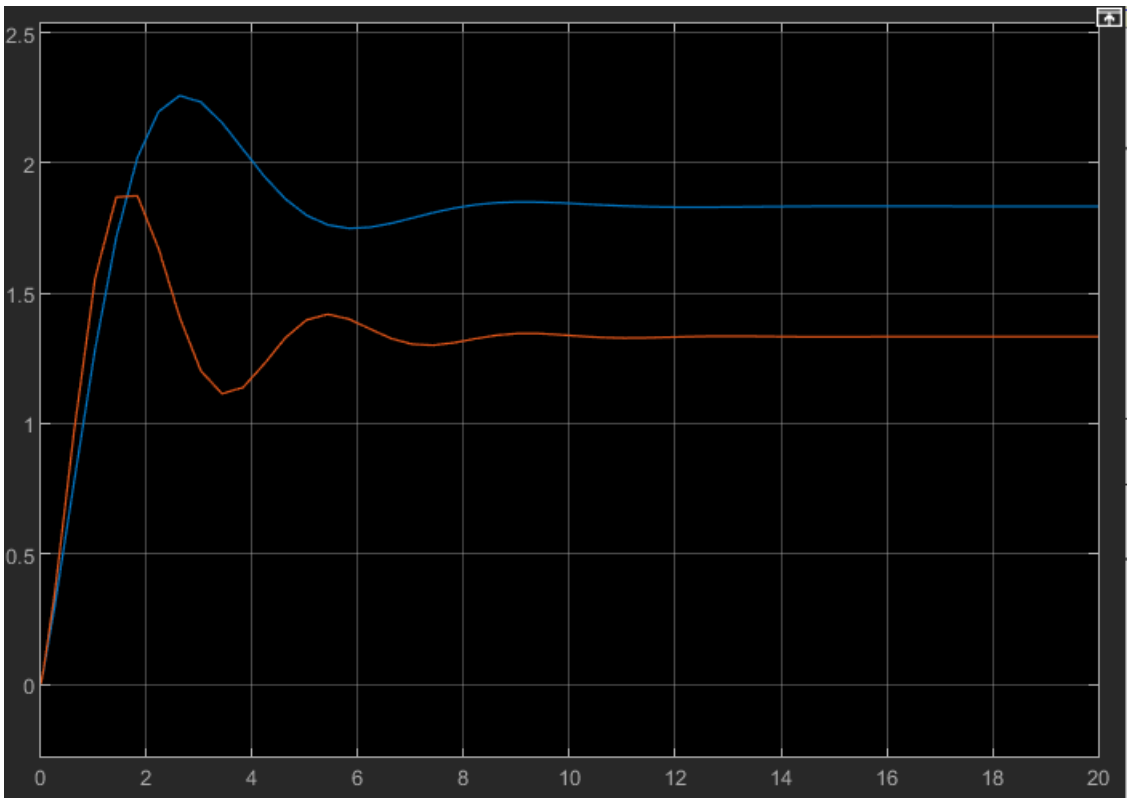
$\boldsymbol{Kp}$ $\boldsymbol{(increase)}$ $\rightarrow$ $Oscillations$ $(increase)$ $\rightarrow$ Mp $(increase)$

$\rightarrow t_d$, t_r , t_p $(decrease)$

$\boldsymbol{Kp}$ $\boldsymbol{(decrease)}$ $\rightarrow$ $Oscillations$ $(decrease)$ $\rightarrow$ Mp $(decrease)$

$\rightarrow t_d$, t_r , t_p $(increase)$

• Varying Kp (n = 1):



Kp = 3
Kp = 1.2
n = 1 (constant)

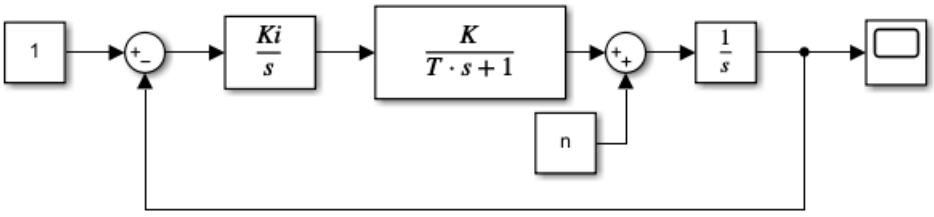
*Notes:

$\boldsymbol{Kp}$ $(\boldsymbol{increase}) \rightarrow Oscillatoins\ (increase) \rightarrow e_{ss}\ (decrease)$

$\boldsymbol{Kp}$ $(\boldsymbol{decrease}) \rightarrow Oscillatoins\ (decrease) \rightarrow e_{ss}\ (increase)$

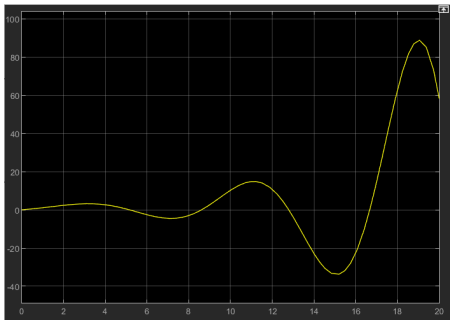
• Effect of increasing noise signal (n) → steady state error [e_{ss}] increases

2. I Controller

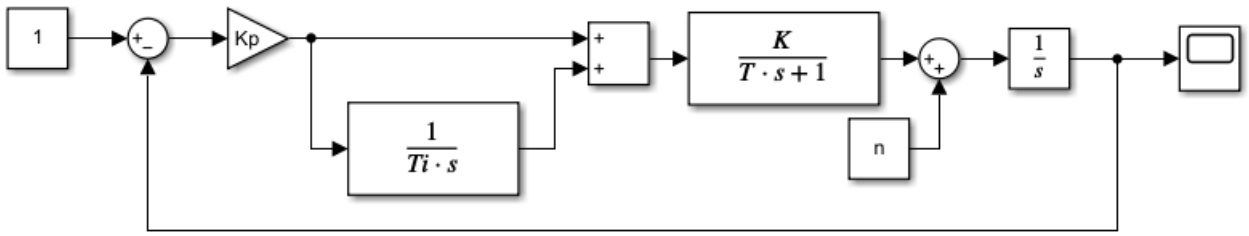


*Notes:

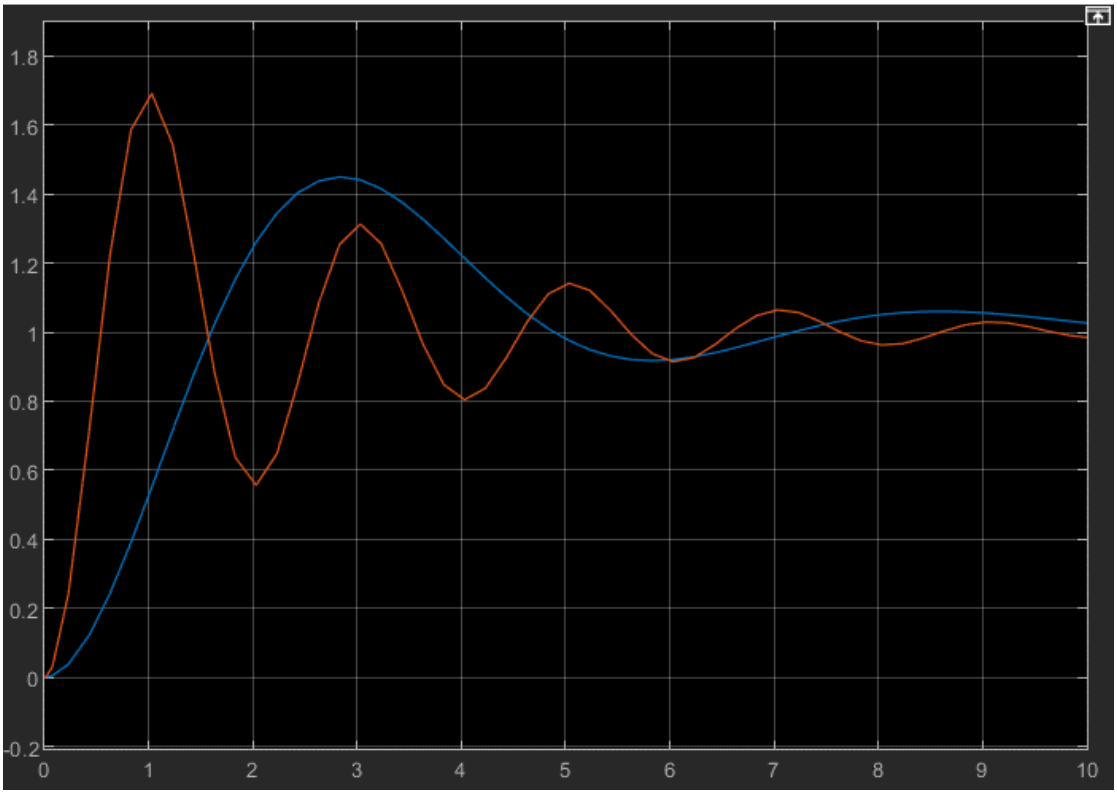
$System\ \boldsymbol{Unstable}\ for\ all\ values\ of\ Ki$



3. PI Controller



• Varying Kp (Ti = Constant, n=0)



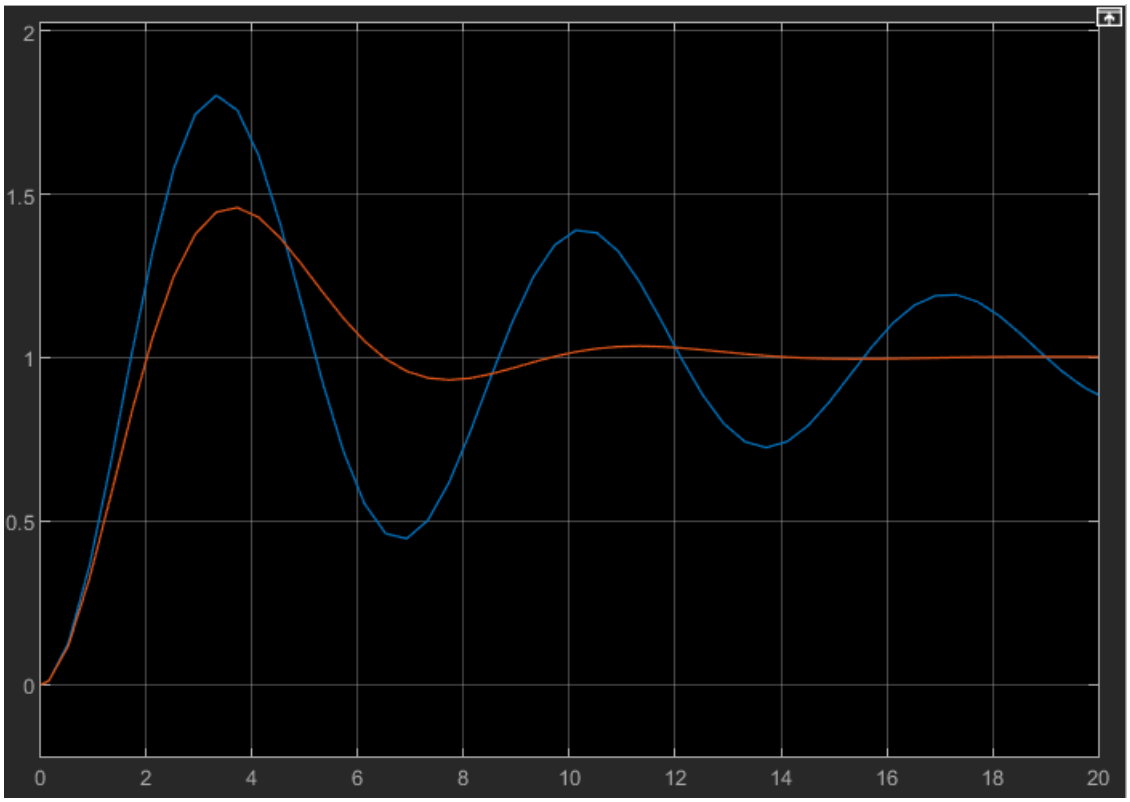
Kp = 10
Kp = 1.5
n=0 (constant)
Ti = 5 (constant)

*Notes:

$$Kp\ (increase) \rightarrow Oscillations\ (increase) \rightarrow Mp\ (increase) \rightarrow t_{ss}\ (increase)$$
$$\rightarrow t_d\ , t_r\ , t_p\ (decrease)$$

$$Kp\ (decrease) \rightarrow Oscillations\ (decrease) \rightarrow Mp\ (decrease) \rightarrow t_{ss}\ (decrease)$$
$$\rightarrow t_d\ , t_r\ , t_p\ (increase)$$

• Varying Ti (Kp = Constant, n=0)



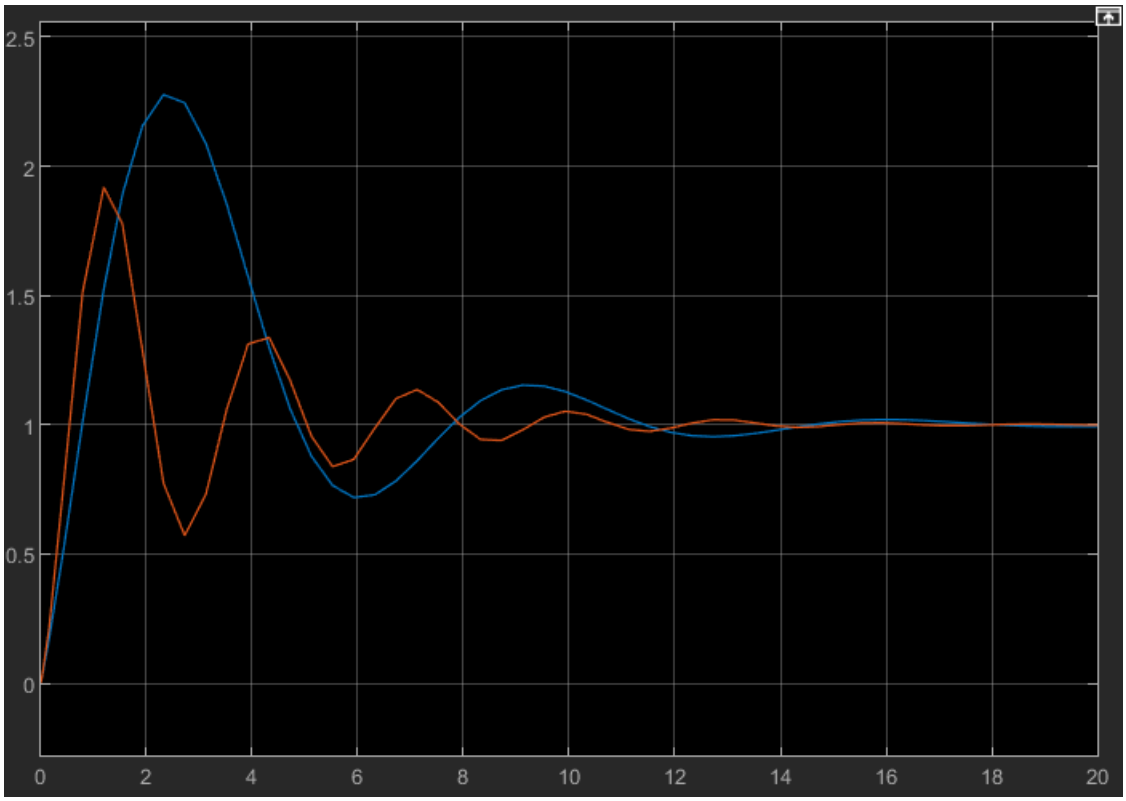
Ti = 4
Ti = 1.5
n=0 (constant)
Kp = 1 (constant)

*Notes:

$Ti\ (increase) \rightarrow Oscillations\ (decrease) \rightarrow t_{ss}\ (decrease) \rightarrow Mp\ (decrease)$

$Ti\ (decrease) \rightarrow Oscillations\ (increase) \rightarrow t_{ss}\ (increase) \rightarrow Mp\ (increase)$

• Varying Kp (Ti = Constant, n = 1)



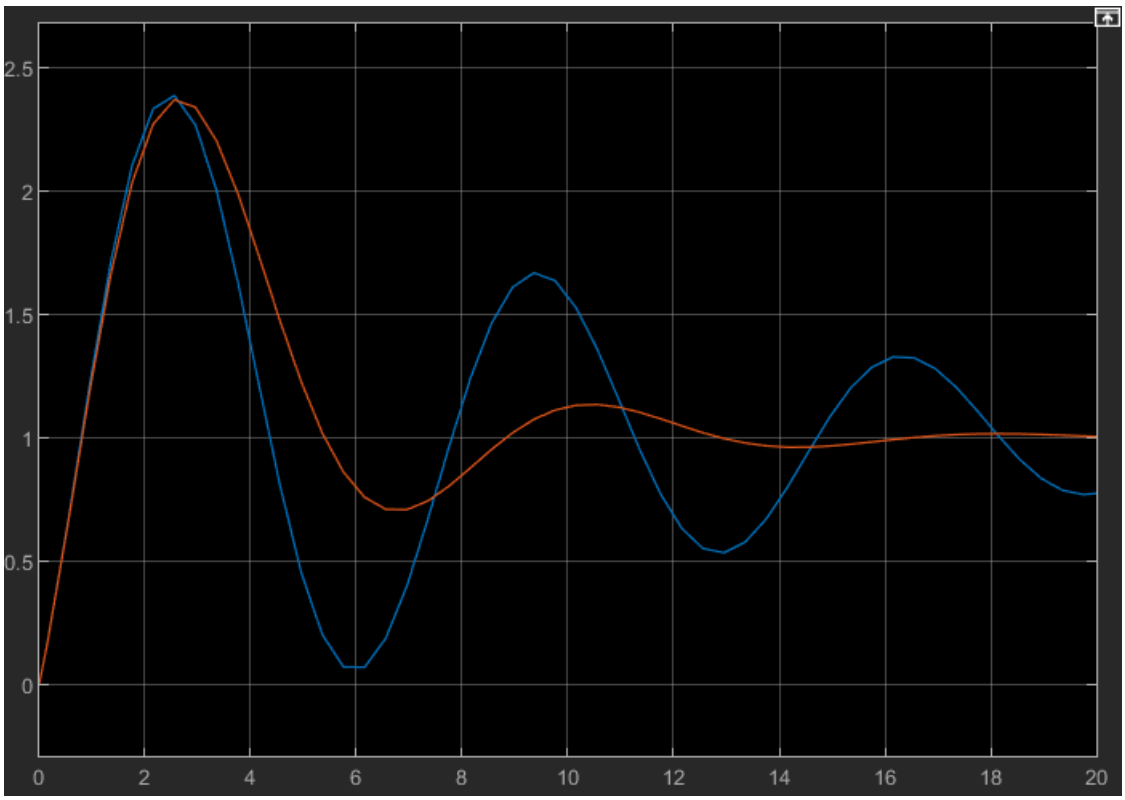
	Kp = 5
	Kp = 1.2
n = 1 (constant)	
Ti = 3 (constant)	

*Notes:

$Kp\ (increase) \rightarrow Oscillations\ (increase) \rightarrow t_{ss}\ (decrease) \rightarrow Mp\ (decrease)$

$Kp\ (decrease) \rightarrow Oscillations\ (decrease) \rightarrow t_{ss}\ (increase) \rightarrow Mp\ (increase)$

• Varying Ti (Kp = Constant, n = 1)



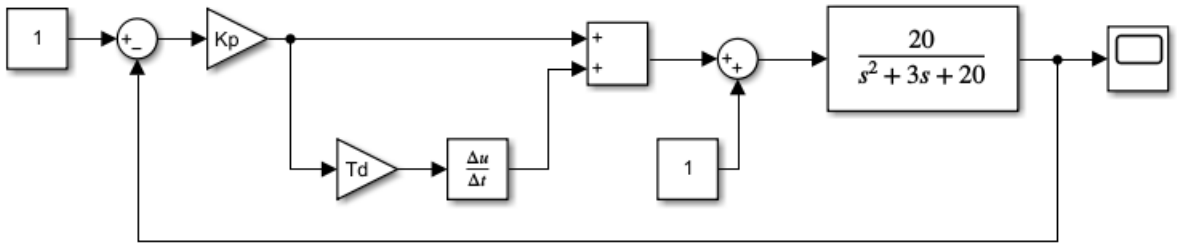
Ti = 3
Ti = 1.5
n = 1 (constant)
Kp = 1 (constant)

*Notes:

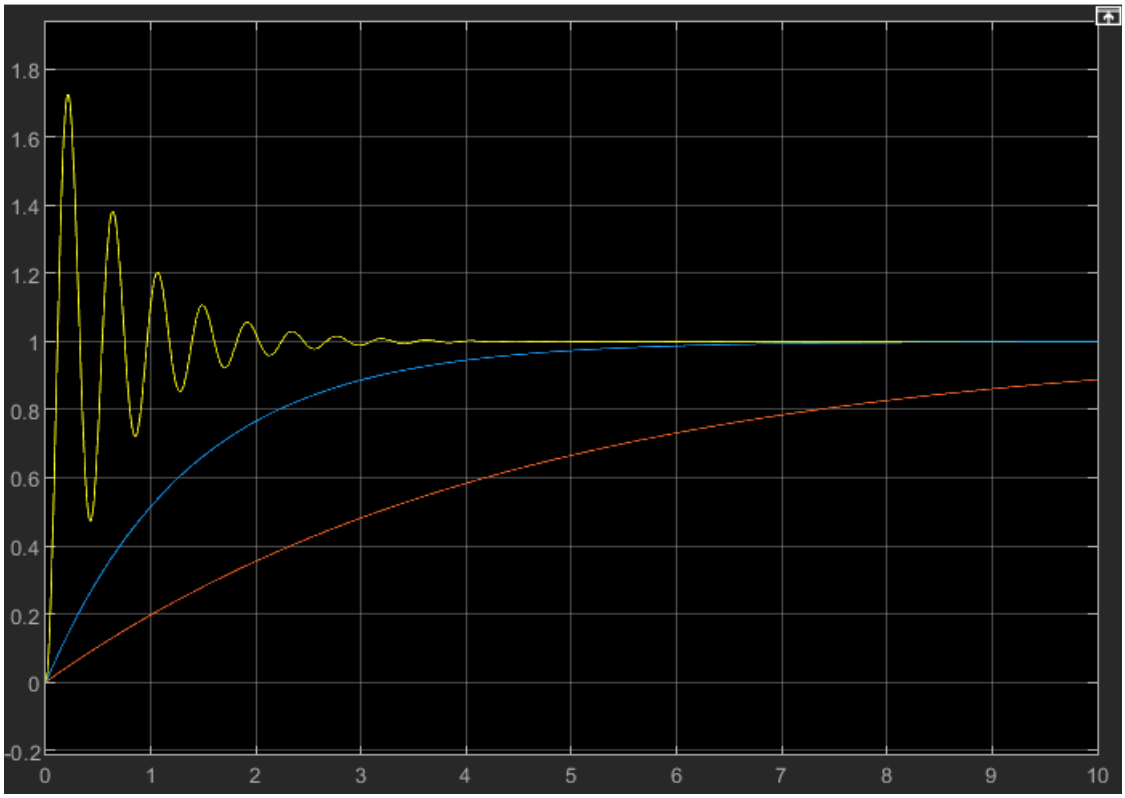
$Ti\ (increase) \rightarrow Oscillations\ (decrease) \rightarrow t_{ss}\ (increase)$

$Ti\ (decrease) \rightarrow Oscillations\ (increase) \rightarrow t_{ss}\ (decrease)$

4. PD Controller



• Varying Td (Kp = constant)



Td = 5
Td = 1.5
Td = 0
Kp = 10 (constant)

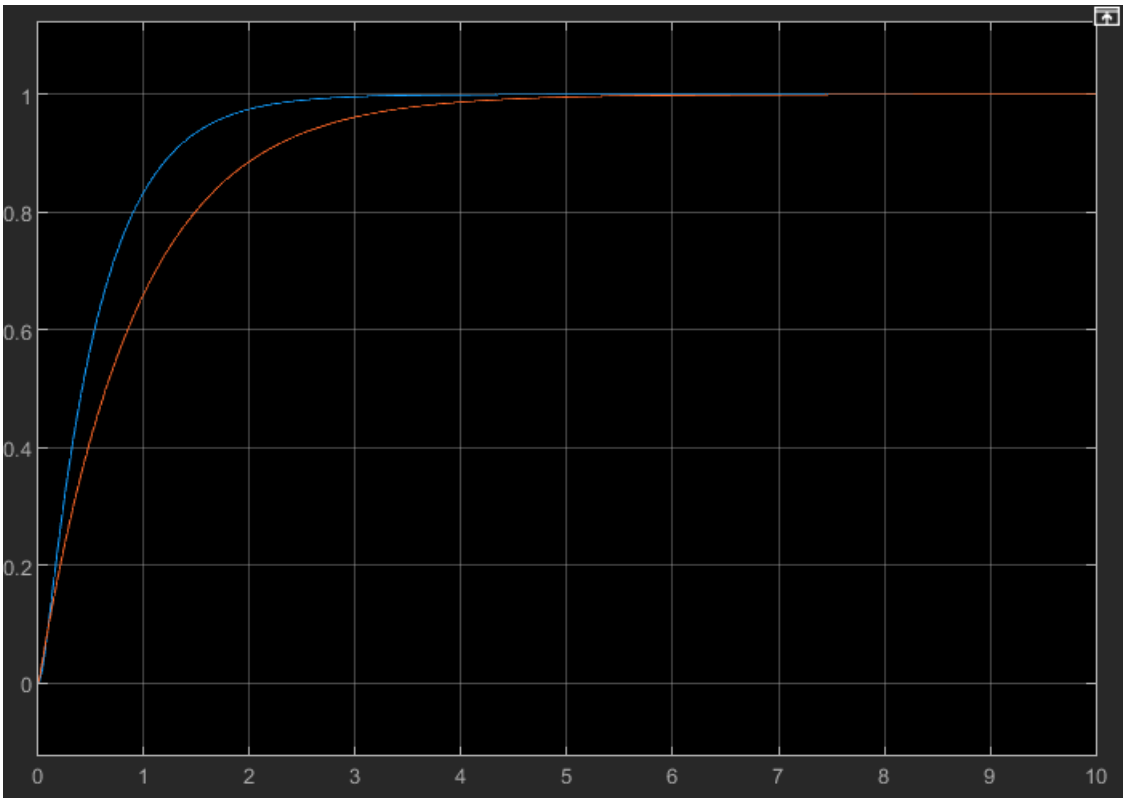
*Notes:

$Adding\ D\ Controller \rightarrow Adds\ Zero\ to\ transfer\ function$
$(oscillations\ and\ overshoot\ minimized)$

$Td\ (increase) \rightarrow t_{ss}\ (increase)$

$Td\ (decrease) \rightarrow t_{ss}\ (decrease)$

• Varying Kp (Td = constant)



	Kp = 10
	Kp = 1
Td = 1 (constant)	

*Notes:

$Kp\ (increase) \rightarrow t_{ss}\ (increase)$

$Kp\ (decrease) \rightarrow t_{ss}\ (decrease)$

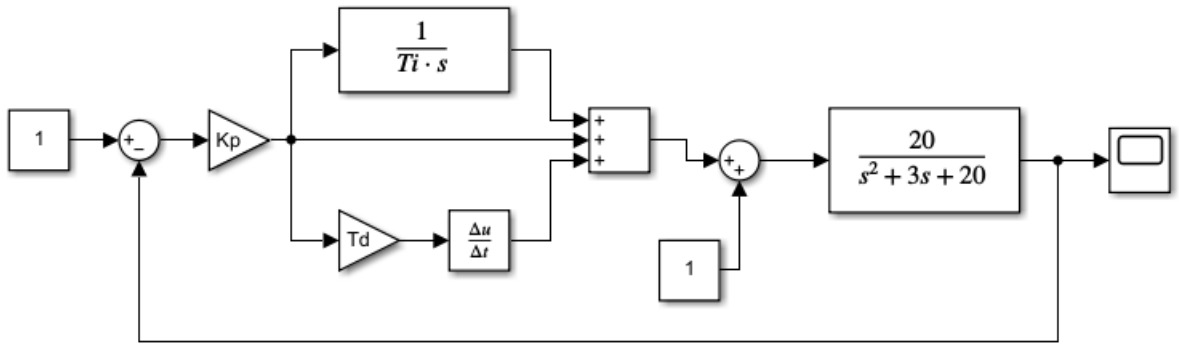
• Why Derivative controller is never used alone?

– Because the magnitude of the controller output is proportional to rate of change of the actuating error signal and not the actuating error itself.

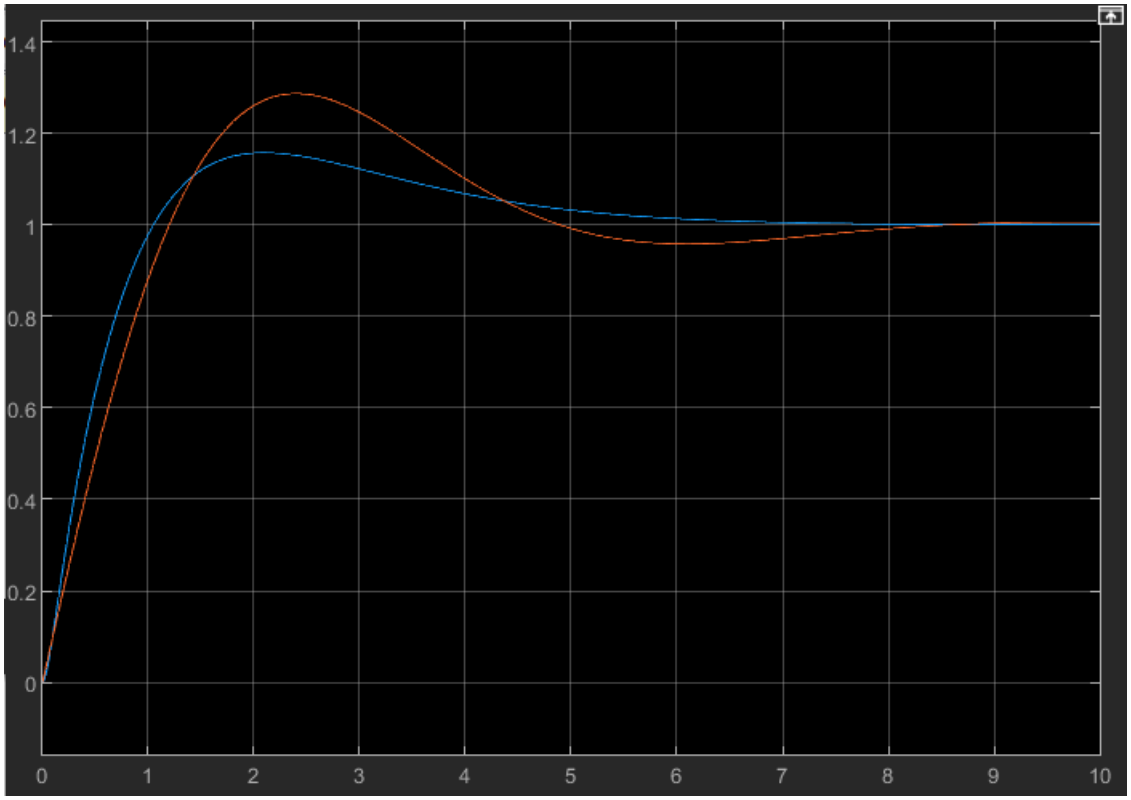
• Derivative control does not affect the steady state error directly, explain how?

– It adds damping to the system and this permits the use of a large value for the gain.

5. PID Controller



• Varying Kp (Ti, Td = constant)



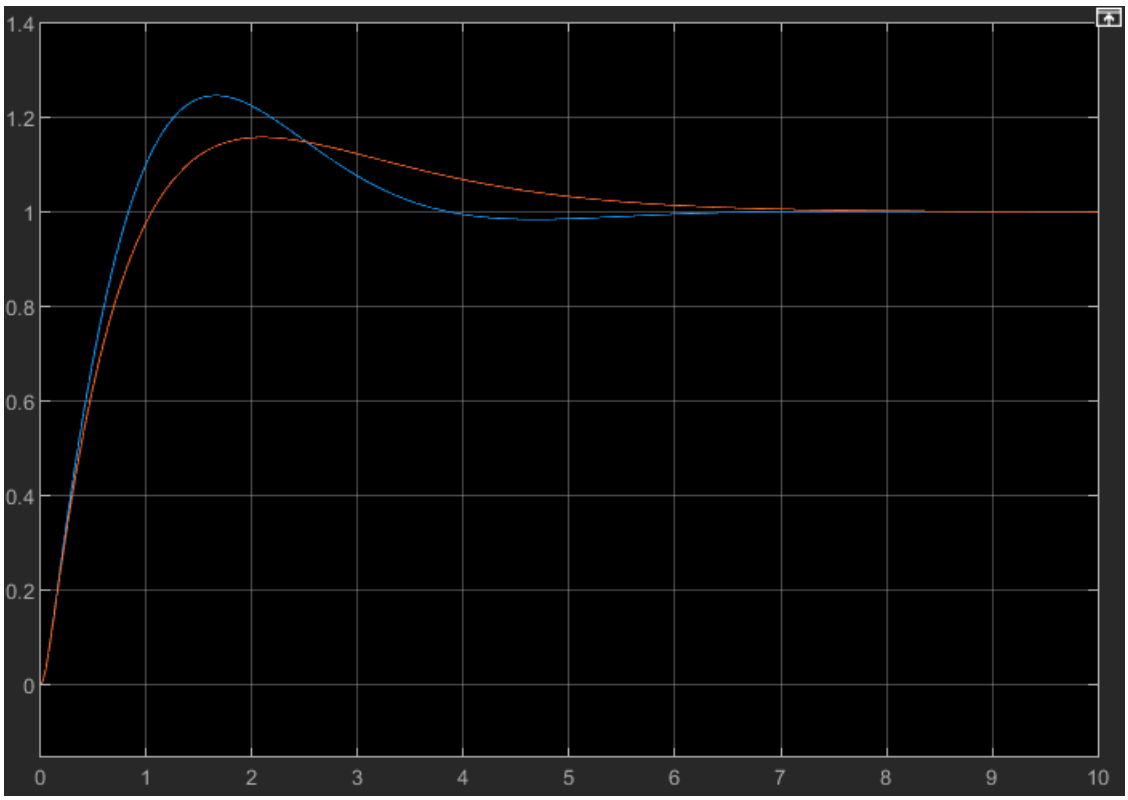
Kp = 10
Kp = 1
Ti, Td = 1 (constant)

*Notes:

$Kp\ (increase) \rightarrow\ t_{ss}\ (increase) \rightarrow Mp\ (increase) \rightarrow Oscillations\ (increase)$

$Kp\ (decrease) \rightarrow t_{ss}\ (decrease) \rightarrow Mp\ (decrease) \rightarrow Oscillations\ (decrease)$

• Varying Ti (Kp, Td = constant)



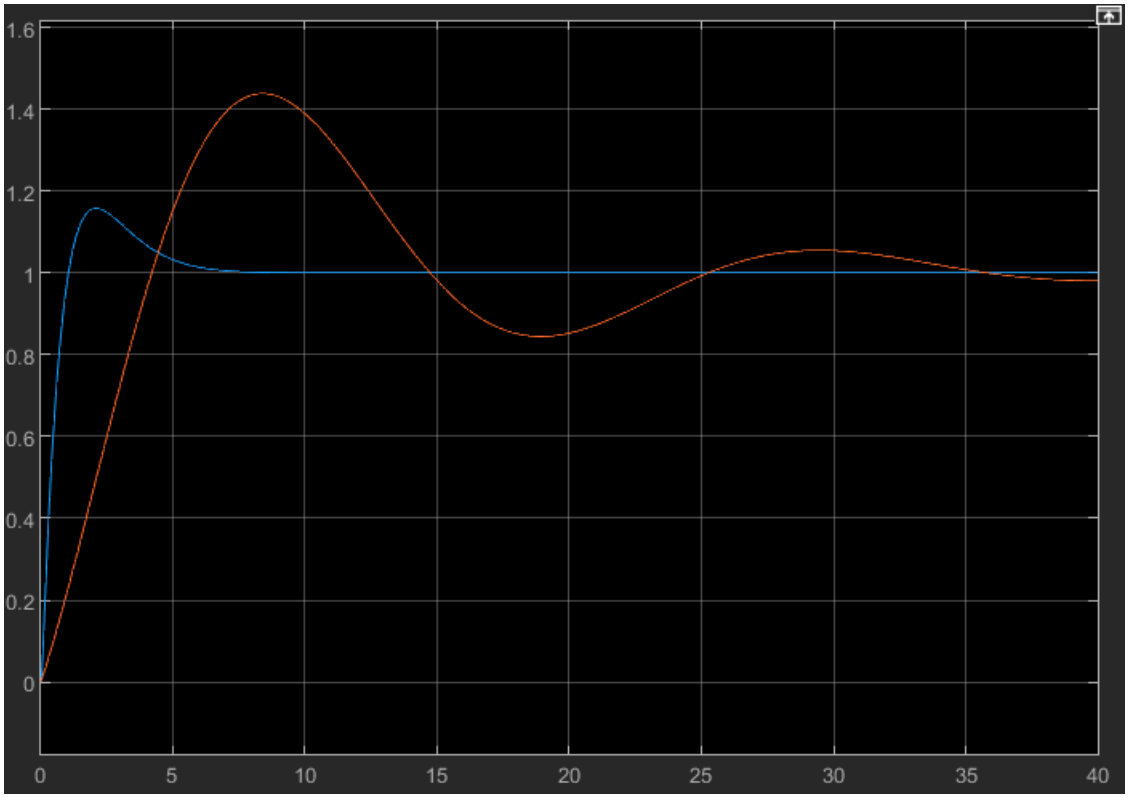
Ti = 1
Ti = 0.5
Kp, Td = 1 (constant)

*Notes:

$Ti\ (increase) \rightarrow\ t_{ss}\ (increase) \rightarrow Mp\ (decrease) \rightarrow Oscillations\ (decrease)$

$Ti\ (decrease) \rightarrow t_{ss}\ (decrease) \rightarrow Mp\ (increase) \rightarrow Oscillations\ (increase)$

• Varying Td (Kp, Ti = constant)



*Notes:

$$Td\ (increase) \rightarrow t_{ss}\ (increase) \rightarrow Oscillations\ (increase) \rightarrow Mp\ (increase) \rightarrow\ t_d, t_r, t_p\ (increase)$$

$$Td\ (decrease) \rightarrow t_{ss}\ (decrease) \rightarrow Oscillations\ (decrease) \rightarrow Mp\ (decrease) \rightarrow\ t_d, t_r, t_p\ (decrease)$$

General Notes:

- **Kp** is termed to the **proportional sensitivity** or the **gain**

- **Td** represents the **derivative time constant**.

- The addition of PD controller will reduce the oscillation of the output and reduce the steady state error.

- The addition of integral control **eliminates** the steady state error.

Experiment (8)

Tuning Rules of PID controller

- Tuning PID:

Adjusting the Parameters of PID controller (Kp, Ti, Td) to meet given performance specifications (Good system response, specific maximum over shoot, etc....)

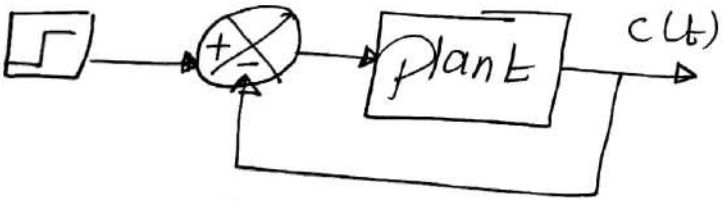
* Ziegler- Nichols both methods aimed to obtain [25% Maximum Overshoot]

- Ziegler- Nichols methods are:

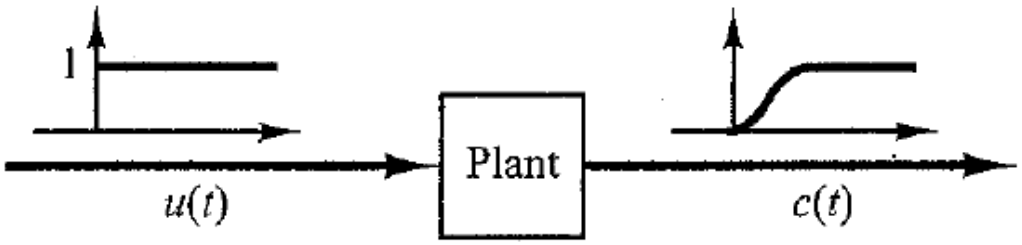
1) Step Response Method [Transfer function is second order]

2) Frequey Response Method [Transfer function is third order]

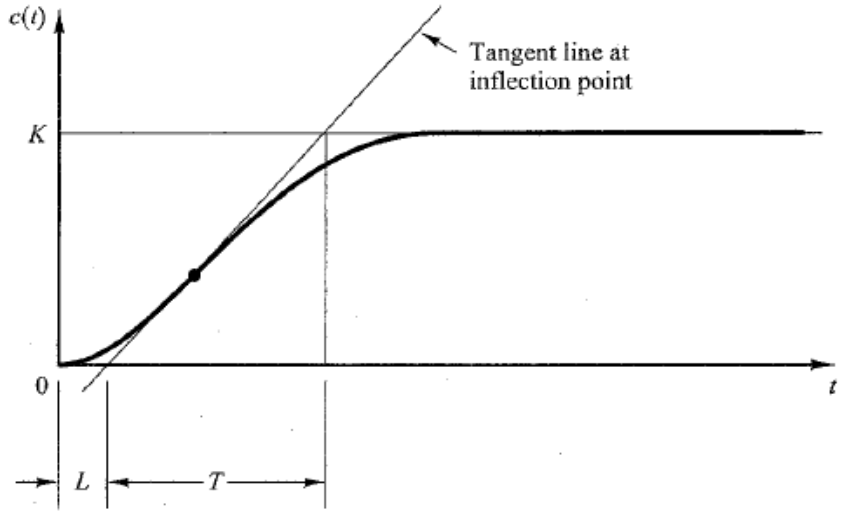
A) First method: step response method (Plant is second order)



- If the plant involves neither integrator(s) nor dominat complex- conjugate poles, the unit step response curve may look like an (S-Shaped curve).



* خطوات إيجاد كل من قيم الثوابت T, L لإيجاد القيم التوافقية لكل من Kp, Ti, Td باستخدام الطريقة الثانية Zieg ler_ Nichols:



1) عند نقطة الالتواء في الموجة، نرسم مماس بحيث يقطع محور الزمن و خط Steady State

2) نحدد قيمة الثابت (L) حيث تكون قيمته النقطة التي يقطع فيها المماس محور الزمن.

3) نحدد قيمة الثابت (T)، حيث تكون فرق القيمة بين الزمن الذي تقطع فيه الموجة خط Steady State وقيمة L

4) نحسب قيم كل من Kp/Ti, Td حسب الجدول أدناه

1) Draw tangent line at inflection point that crosses times axis and steady state line

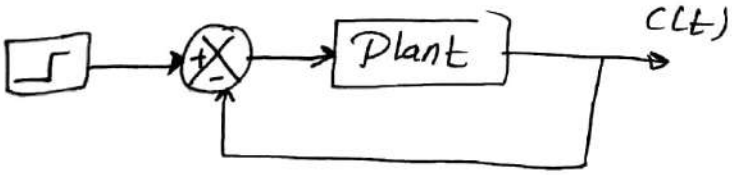
2) Finding L constant, where it is the value of the point on time axis that the tangent line comes a cross.

3) Finding T constant, which is the difference between the point on time axis where the tangent comes across the steady state line and between L value

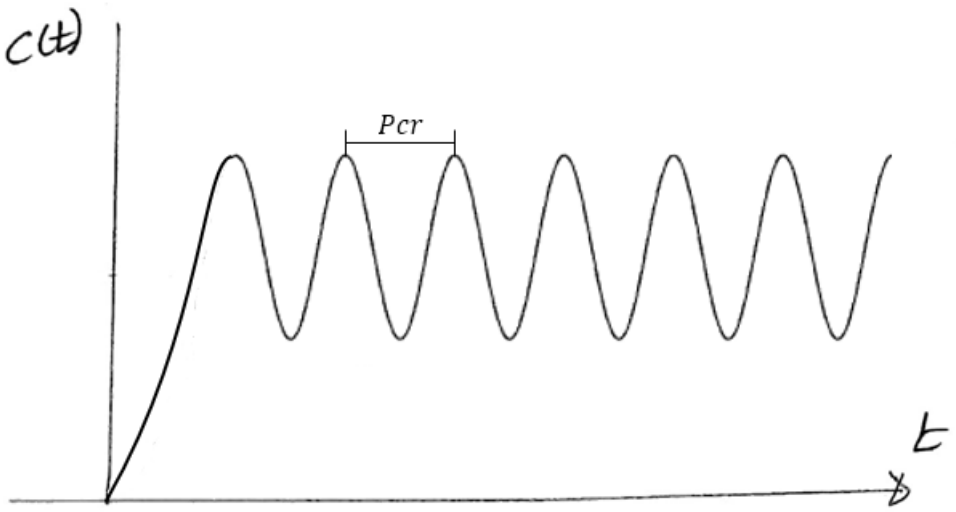
4) Calculating Kp/Ti/Tp depending on the table below

type of controller	Kp	Ti	Td
P	T/L	∞	0
PI	0.9T/L	L/0.3	0
PID	1.2T/L	2L	0.5L

B) Second method: Frequecy response (plant is Third order)



* خطوات ايجاد قيم كل من الثوابت Kcr, Pcr لحساب القيم التوافقية لكل من Kp/Ti/Td باستخدام الطريقة الثانية Ziegler-Nichols:



1) نجعل قيمة المعاملات $Ti=\infty$ و $Td=0$

2) نقوم بتغير قيمة (kp) تدريجيا من الصفر الى قيمة حرجة Kcr عندها الموجة تتعرض لذبذبات مستمرة.

3) نقوم بحساب الفترة الزمنية ما بين قمتين في الموجة التي تسمى الفترة الزمنية الحرجة (Pcr).

4) نحسب قيم كل من kp / Ti / Td حسب الجدول أدناه.

1) We set the parameters $Ti=\infty$ and $Td=0$

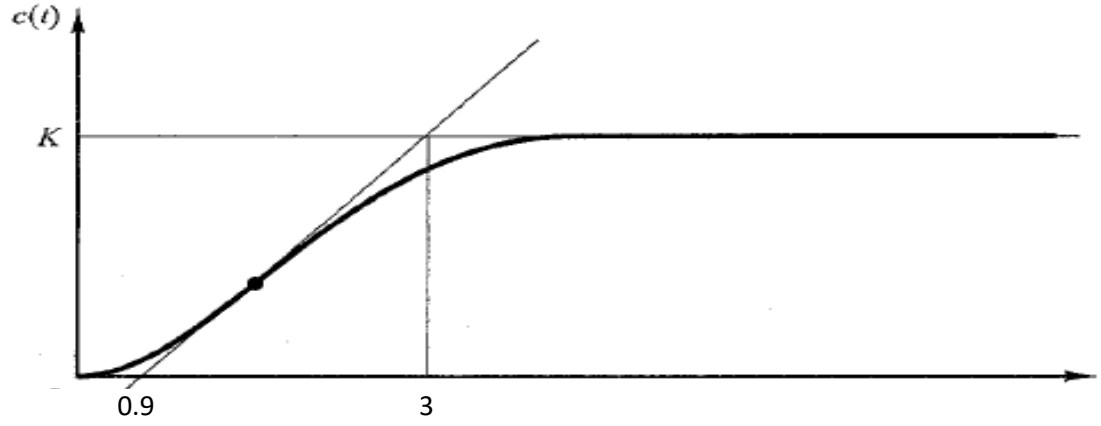
2) Increasing kp from zero to a critical value (Kcr) where the output first exhibits sustained oscillations.

3) Measure the period between two peeks which is called the critical period (Pcr)

4) Calculating KP / Ti / Td depending on the table below

type of controller	kp	Ti	Td
P	0.5 kcr	∞	0
PI	0.45 kcr	Pcr/1.2	0
PID	0.6 kcr	0.5 Pcr	0.125 Pcr

Ex1°- Find the suitable parameters of PI controller depending on the graph below



.Solution

$L = 0.9$

$T = 3 - 0.9 = 2.1$

Parameters of PI controller

$Kp = 0.9 \frac{T}{L} \rightarrow kp = 2.1$

$Ti = \frac{L}{0.3} = 3 \rightarrow Ti = 3$

$Td = 0$

Experiment (9)

Frequency Response Analysis

- Frequency response: the steady state response of control system to a sinusoidal input.

The frequency response characteristics can be obtained directly from the sinusoidal transfer function by replacing (s) in the transfer function G(s) with (jw) to get sinusoidal transfer function G(jw).

* G(jw) is complex quantity can be written as:

$G(jw) = |G(jw)| \cdot e^{j\phi}$ or $|G(jw)| \angle \phi$

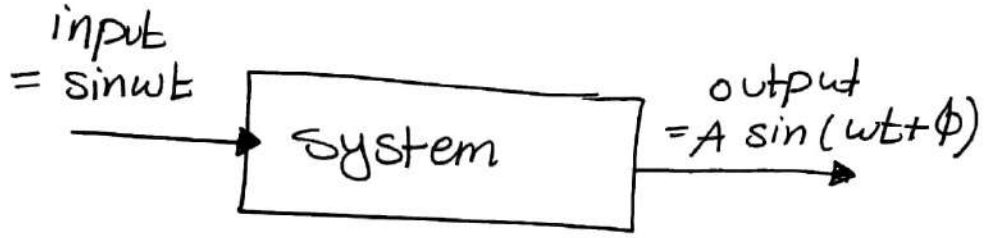
$|G(jw)| \longrightarrow$ represent the magnitude

$\phi \longrightarrow$ represent the Phase angle

when we have a sinusoidal input the output signal will be sinusoidal of the same freqeucy of the input.

The Amplitude of the output is given by the Product of the input and the magnitude $|G(jw)|$

— The Phase angle differs from that of input by amount (ϕ)



- Amplitude = $A = |input| \times |G(jw)|$.
- Frequency = w = (remains the same)
- Phase shift occurs by amount (ϕ).

* Remember !!

* $j = 1 \angle 90°$

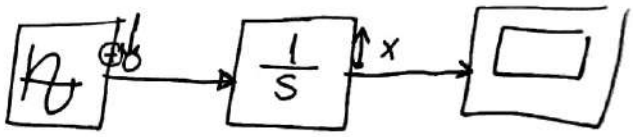
* $\frac{1}{j} = -j = 1 \angle -90°$

* $j \rightarrow 0 + 1j \rightarrow \tan^{-1}\left(\frac{1}{0}\right) \rightarrow \tan^{-1}(\infty) = 90°$

* $j \rightarrow 0 - 1j \rightarrow \tan^{-1}\left(\frac{-1}{0}\right) \rightarrow \tan^{-1}(-\infty) = -90°$

* $j^2 \rightarrow 1 \angle 90° * 1 \angle 90° \rightarrow 1 \angle 180° = -1$

A) Integral Factor:



Transfer function $G(s) = \frac{1}{s}$

Sinusoidal Transfer function $G(s) = \frac{1}{jw}$

$|G(jw)| = \frac{1}{w}$ $\quad \phi = -90$

Logarithmic Magnitude $(Lm) = 20\log(|G(jw)|)$

$= 20\log(\frac{1}{w})$

$= -20\log(w)$

B) Derivative Factor



Transfer function $G(s) = s$

Sinusoidal Transfer function $G(jw) = jw$

$|G(jw)| = w$ $\quad \phi = +90^\circ$

Logarithmic Magnitude $(Lm) = 20\log(|G(jw)|$

$= 20\log(w)$

c) First order Factor



Transfer Function

$$G(s) = \frac{K}{1+sT}$$

Sinusoidal Transfer Function $G(j\omega) = \frac{K}{1+j\omega T}$

$$|G(j\omega)| = \left|\frac{K}{1+j\omega T}\right| = \frac{\sqrt{K^2}}{\sqrt{1+(\omega T)^2}} = \frac{K}{\sqrt{1+(\omega T)^2}}$$

Logarithmic Magnitude (Lm) $= 20\log(|G(j\omega)|)$

$$= 20\log\left(\frac{K}{\sqrt{1+(\omega T)^2}}\right)$$

$$= 20\left[\log(k) - \log\sqrt{1+(\omega T)^2}\right]$$

cutt of frequency (ω_c) $\rightarrow \omega = \frac{1}{T}$

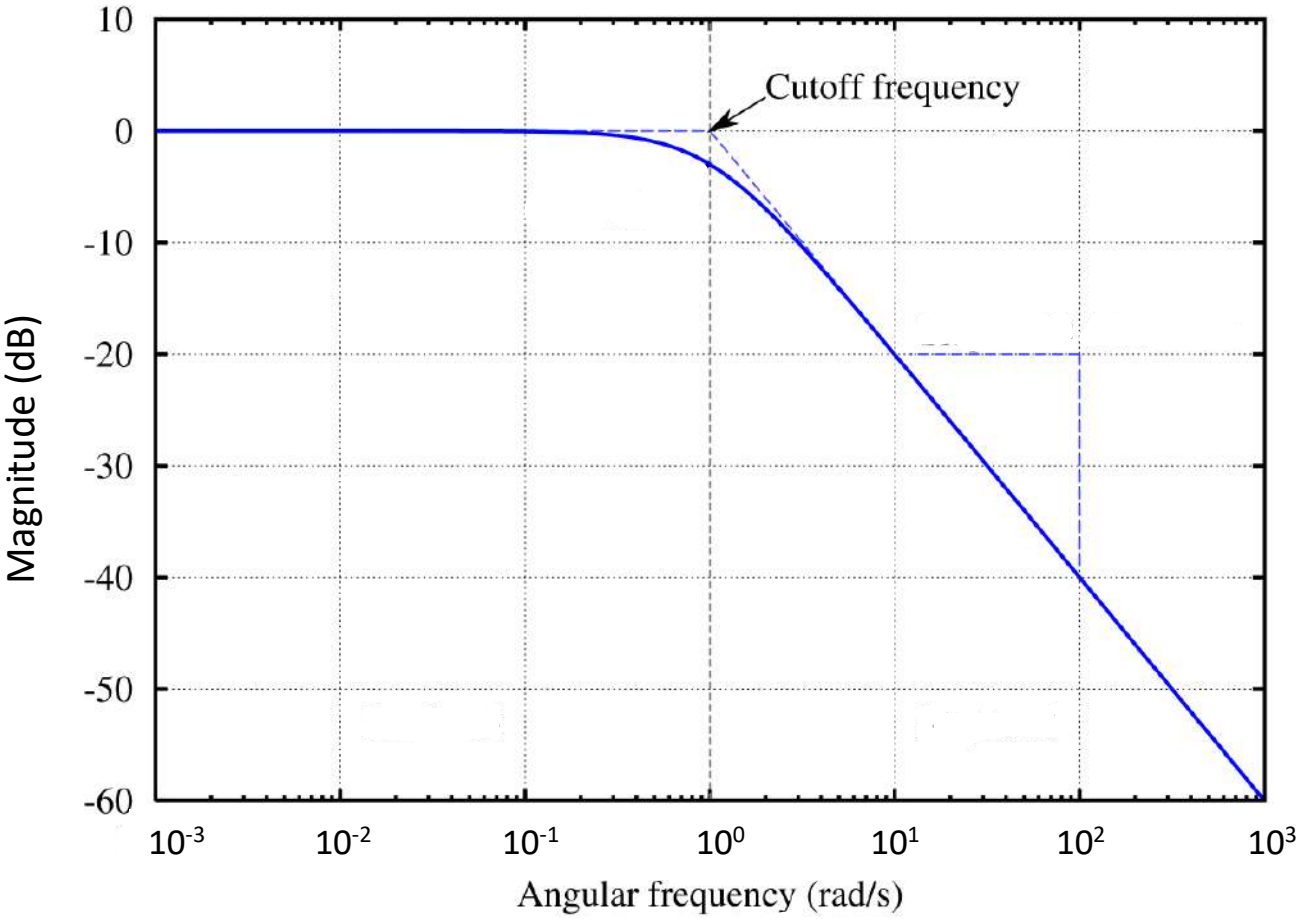
when $\omega = \frac{1}{T} \longrightarrow \phi = -\tan^{-1}(\omega T) = -\tan^{-1}\left(\frac{1}{T}\cdot T\right) =$

$-\tan^{-1}(-1) = -45$

cutt of frequency occurs at ($\omega = \frac{1}{T}$) and phase angle $\phi = -45°$.

* Bodeplot (First order)

Cutt off frequency (ω_c): the frequency at which the two asymptotes cuts or meet each other, also known as corner frequency or break frequency.



* أثر كل من T / k على شكل Bode plot في رسم

- $L m_{max}$ ⟶ when $(w=0)$ ⟶ $Lm_{max}|_{w=0} = 20[\log(k) - \log(1)]$
- $L m_{max} = 20 \log(k)$
- $w_c = \frac{1}{T}$

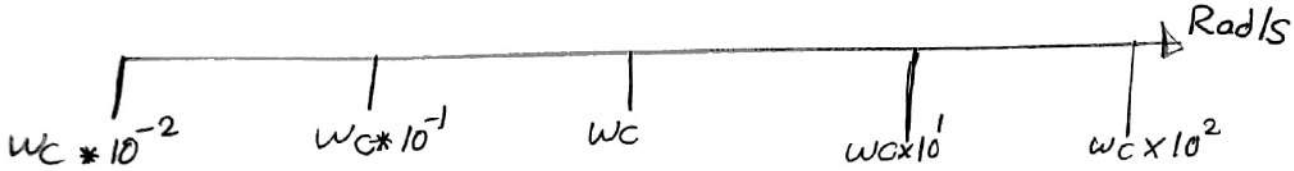
* k (increase) ⟶ Lm_{max} (increase)
* k (decrease) ⟶ Lm_{max} (decrease)
* T (increase) ⟶ f_c (decrease)
* T (increase) ⟶ f_c (increase)

* خطوات رسم Bode plot لنظام First order :-

① نجد قيمة Lm_{max}

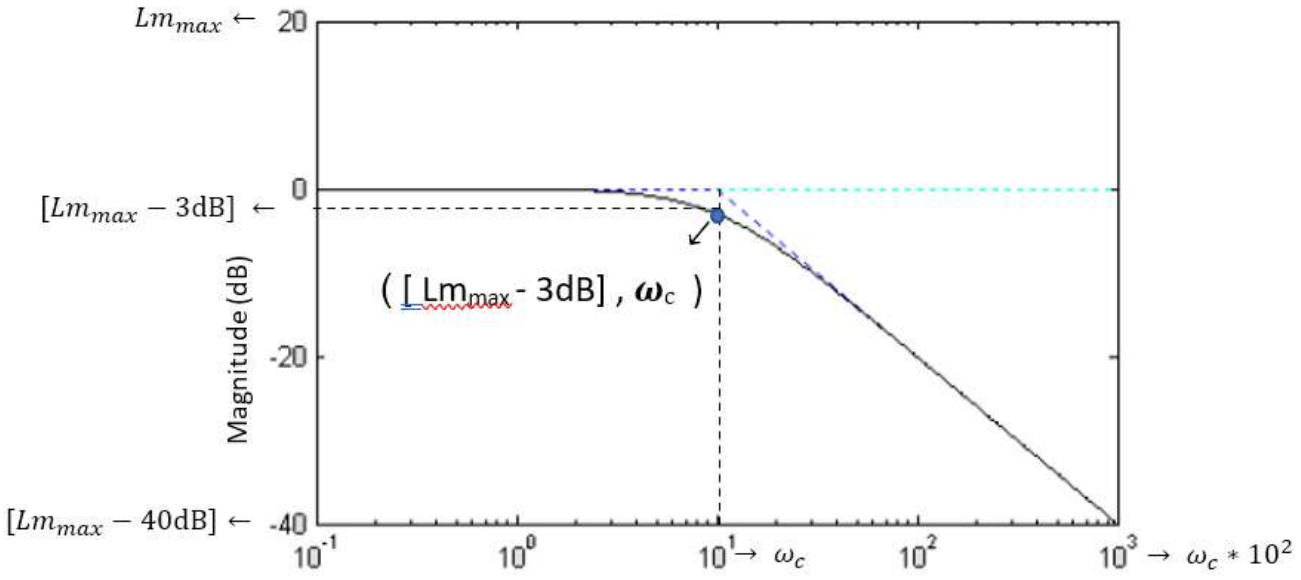
② نجد قيمة ω_c

③ نقوم بتقسيم محور التردد إلى خمسة أجزاء على النحو التالي:

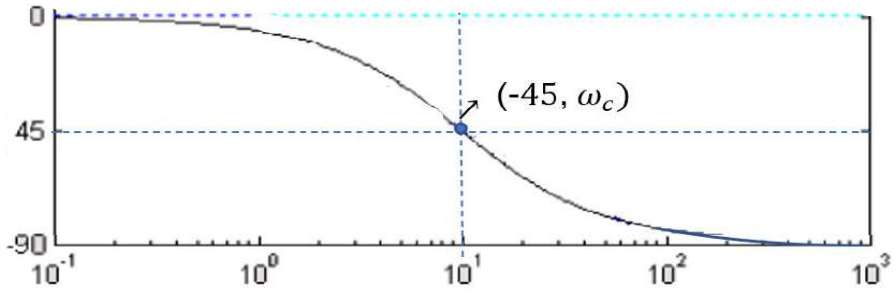


④ لرسم منحنى Magnitude (dB) نقوم برسم خط مستقيم موازي لقيمة Lm_{max} وخط مستقيم عامودي عند ω_c ومن نقطة تقاطع الخطين نقوم برسم خط مائل مستقيم آخر يمر بالنقطة $[[Lm_{max} - 40dB], \omega_c \times 10^2]$ نقوم برسم خط أفقي على طول إلى الخط الموازي لقيمة Lm_{max} بقليل ثم نرسم انحناء يمر بالنقطة $[[Lm_{max} - 3dB], \omega_c)$

ثم نصله بعدها بالخط المائل لنحصل على شكل قريب من الشكل أدناه.



5) لرسم منحنى Phase (deg) نرسم رسمة تشبه الاقتران التكعيبي ($Y = X^3$) ولكنه مقلوب إلى اليمين بحيث تكون نقطة المركز (-45 , ω_c) ونقطة كما في الشكل أدناه.



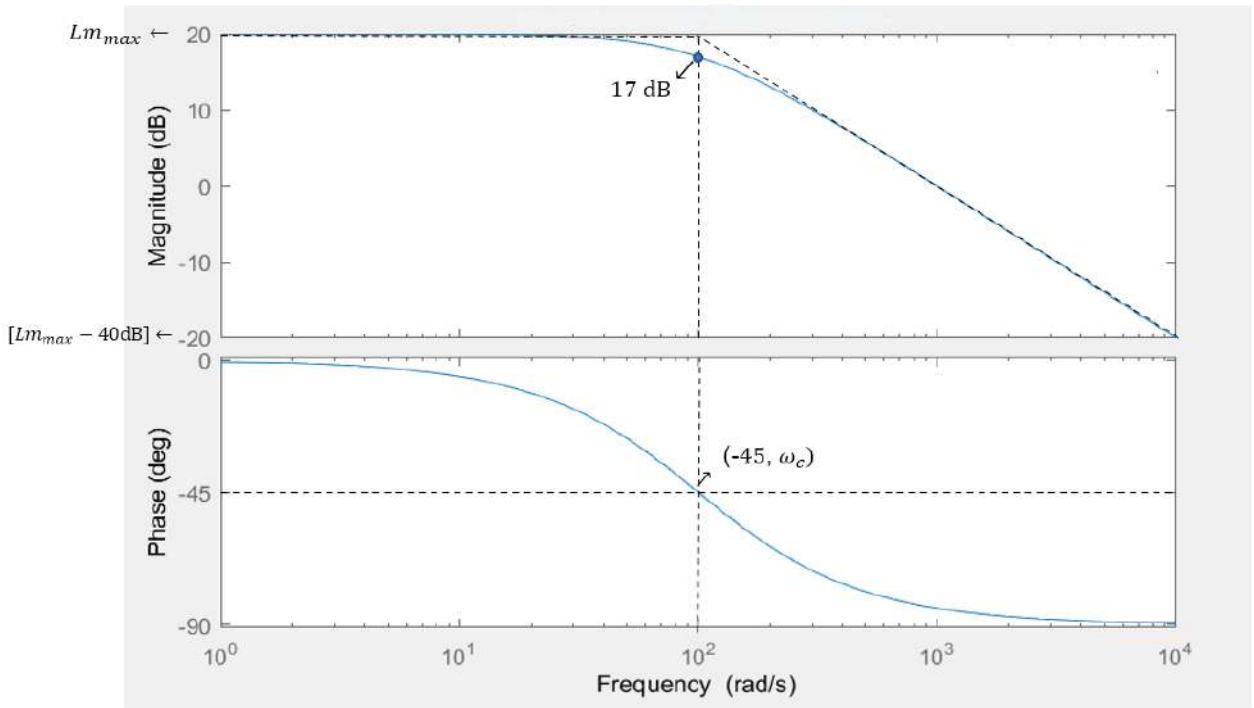
EX1 :- Draw the magnitude and Phase bode plots for the following system:

$$G(S) = \frac{10}{0.01S+1}$$

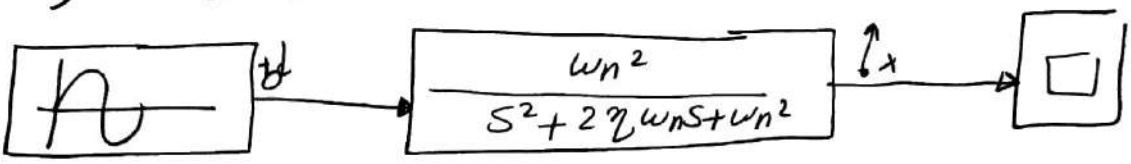
Solution:-

$$Lm_{max} = 20\log(K) = 20\log(10) = 20dB$$

$$\omega_c = \frac{1}{T} = \frac{1}{0.01} = 10^2 \text{ Rad/S}$$



d) Second order Factor



- Transfer function $G(s) = \dfrac{\omega_n^2}{s^2+2\eta\omega_n s+\omega_n^2}$

- Sinusoidal Transfer Function

$$G(s) = \frac{1}{1-\left(\frac{\omega}{\omega_n}\right)^2 + j\,2\eta\frac{\omega}{\omega_n}}$$

$$\phi = -\tan^{-1}\left(\frac{2\xi(\omega/\omega_n)}{1-(\omega/\omega_n)^2}\right)$$

Logarithmic magnitude $(Lm) = -20\log\left(\sqrt{\left(1-\frac{\omega^2}{\omega_n^2}\right)^2+\left(2\eta\frac{\omega}{\omega_n}\right)^2}\right)$

- Resonance Frequency $\longrightarrow \omega = \omega_n$

when $(\omega=\omega_n) \longrightarrow \phi = \tan^{-1}\left(\frac{2\eta(\omega/\omega_n)}{1-\left(\frac{\omega}{\omega_n}\right)^2}\right) =$

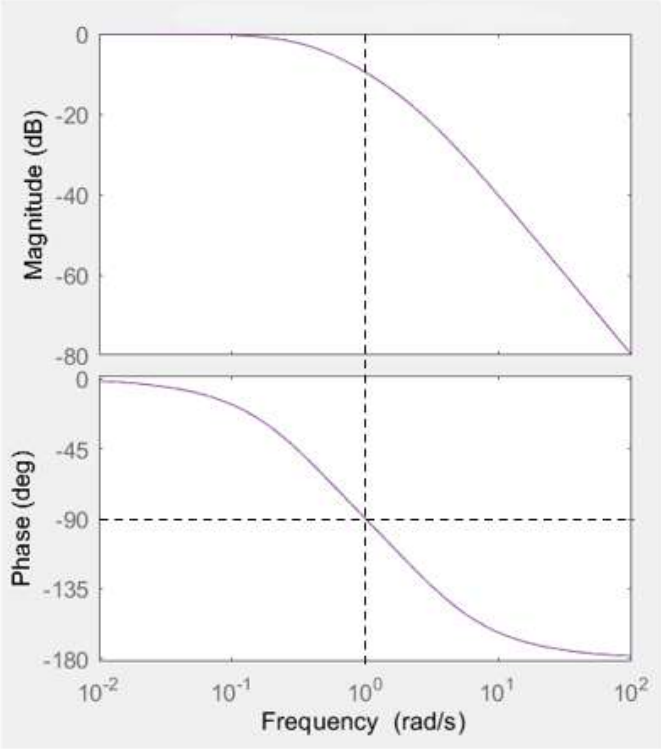
$$-\tan^{-1}(\infty) = -90^\circ$$

Resonance Frequency Occurs at $(\omega=\omega_n)$ and

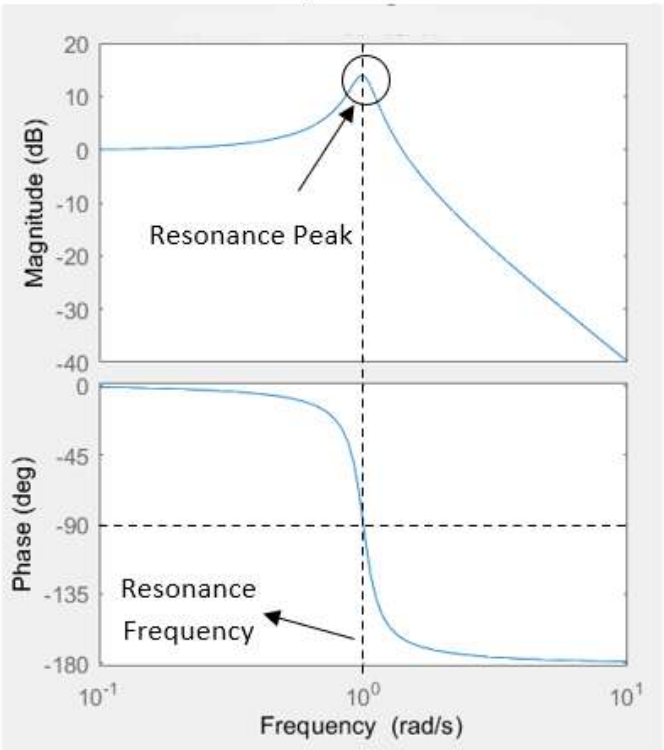
Phase angle $\phi = -90^\circ$

أثر (ζ) Damping Ratio:

$\zeta = 1.5$



$\zeta = 0.1$



ζ (increase) $\rightarrow$ *Resonance Peak* (*increase*)

$\rightarrow$ *Phase change at resonance frequency*(*decrease*)

ζ (decrease) $\rightarrow$ *Resonance Peak* (*decrease*)

Final Exam
(Experiment 6, 7, 8, 9)

Q.1: Choose the correct answer: [24 Marks]

1. In frequency response, the steady state phase angle for first order is:

a. 90˚ b. 180˚

c. -90˚ d. -180˚

2. In frequency response, the steady state phase angle for second order is:

a. 90˚ b. 180˚

c. -90˚ d. -180˚

3. Decreasing the gain (Kp) in first order system with P-controller:

a. increase the steady state value. b. decrease the error.

c. increase settling time. d. decrease steady state value.

4. In Ziegler – Nichols second method, we keep increase (Kp) from zero to (Kcr) value where the response becomes:

a. Overdamped. b. Underdamped.

c. Unstable. d. Undamped.

5. Decreasing the natural frequency ($\boldsymbol{\omega}_n$) in second order system:

a. increase peak time. b. decrease settling time.

c. decrease error d. increase maximum overshoot

6. Ziegler – Nichols PID Tuning methods aim to obtain:

a. 0% maximum overshoot. b. 10% maximum overshoot.

c. 25% maximum overshoot. d. 20% maximum overshoot.

7. In frequency response, the maximum logarithmic value is obtained when:

a. T = 0 b. $\boldsymbol{\omega} = \infty$

c. T = ∞ d. $\boldsymbol{\omega} = 0$

8. The sinusoidal transfer function for control system where x is input, y is output:

a. X(s)/Y(s)

b. Y(jω)/X(jω)

c. Y(s)/X(s)

d. X(jω)/Y(jω)

9. If the damping ratio (ζ = -0.2), the system response is:

a. Undamped.

b. Critically damped.

c. Unstable.

d. Underdamped.

10. Increasing (Td) in second order system with PID - Controller:

a. decrease oscillations.

b. increase maximum overshoot.

c. unstable for all (Td) values.

d. decrease settling time.

11. The main use for I - Controller in PID - Controller is to:

a. decrease error.

b. decrease oscillations.

c. to lower the order of the system.

d. eliminating steady state error.

12. For a second order system, if the response was Overdamped, one of the following values is the damping ratio of the system:

a. $\zeta = 0$

b. $\zeta = 0.707$

c. $\zeta = 1.707$

d. $\zeta = 1$

13. Ziegler – Nichols second method has been used on a system, the transfer function of the system was:

a. $G(s) = \frac{1}{3s+1}$

b. $G(s) = \frac{4}{s^2+6s+4}$

c. $G(s) = \frac{72}{s^3+20s^2+124s+24}$

d. $G(s) = \frac{1}{s}$

14. For second order system, if $\zeta = 0.5$, $t_{ss(2\%)} = 8s$, ω_n equals:

a. $\omega_n = 1$

b. $\omega_n = 0.75$

c. $\omega_n = 1.25$

d. $\omega_n = 0.39$

15. For $G(s) = \frac{1}{(s+1)(s+4)}$, we use Ziegler – Nichols':

a. first method.

b. second method.

c. we use another tuning method.

d. we don't use any method.

16. In frequency response, resonance frequency happens when:

a. $\boldsymbol{\omega} = \boldsymbol{\omega}_n$

b. $\boldsymbol{\omega} = \frac{1}{T}$

c. $\boldsymbol{\omega} = \boldsymbol{\omega}_c$

d. $Lm_{max} = 20 \log (K)$

17. One of the following should be considered when using D controller:

a. Td value should be as high as possible.

b. Td value should be as low as possible.

c. using P controller with it.

d. Choosing Td value based on tuning method.

18. Given the damping ratio ($\boldsymbol{\zeta} = 0.3$), natural frequency ($\boldsymbol{\omega}_n = 3.1448$), the rise time ($t_r$) equals:

a. -23.13 s

b. 0.625 s

c. -25.82 s

d. 23.13 s

Q.2: Based on the graph of an output signal of a system answer the following: [12 Marks]

a. Rise time =

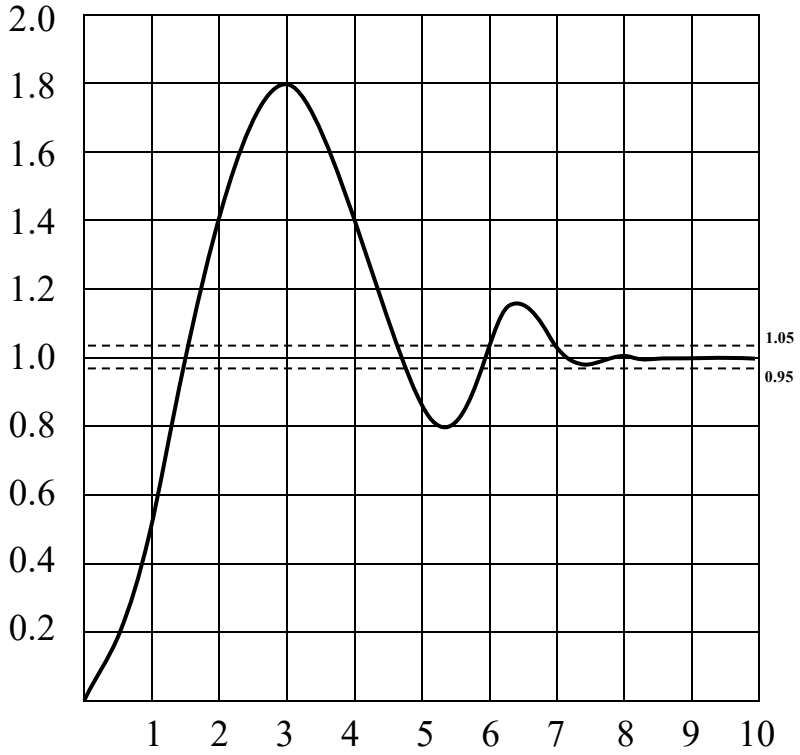
b. Peak time =

c. Delay time =

d. Maximum Overshoot =

e. Type of Response =

f. Steady State Time (2% Criterion) =



Q.3: [14 Marks]

a. For Ziegler – Nichols second method, write down the steps of finding Kcr, Pcr values for PID controller parameters tuning. [4 Marks]

b. For a Plant, we drew a tangent line at inflection point and the constants T and L respectively were T = 1.8, L = 0.3, using these values fill the table below: [9 Marks]

Type of Controller	Kp	Ti	Td
P			
PI			
PID			

c. When used Ziegler – Nichols second method to tune a PID controller, P_{cr} value was 2.5 sec, (Ti) value should be _____ . [1 Mark]

Final Exam Answers

Q.1:

1. C	2. D	3. D	4. D	5. A	6. C
7. D	8. B	9. C	10. B	11. D	12. C
13. C	14. A	15. A	16. A	17. C	18. B

Q.2:

a. Rise time = 1.5 s

b. Peak time = 3 s

c. Delay time = 1 s

d. Maximum Overshoot = 1.8 – 1 = 0.8

e. Type of Response = Underdamped

f. Steady State Time $_{(2\% \text{ Criterion})}$ = 7 s

Q.3:

a.

1. We set the parameters Ti = ∞ and Td = 0

2. Increasing Kp from zero to a critical value (Kcr) where the output first exhibits sustained oscillations.

3. Measure the period between two peeks which is called the critical period (Pcr).

4 .Calculating Kp, Ti, Td depending on the table below:

Type of Controller	Kp	Ti	Td
P	0.5 Kcr	∞	0
PI	0.45 Kcr	Pcr / 1.2	0
PID	0.6 Kcr	0.5 Pcr	0.125 Pcr

b.

Type of Controller	Kp	Ti	Td
P	6	∞	0
PI	5.4	1	0
PID	7.2	0.6	0.15

c. Ti = 5